Master Geomechanics, Civil Engineering and Risks

Numerical study of the thermo-hydro-mechanical behavior of concrete under severe accident conditions using a stochastic poro-mechanical approach

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Finally, I would like to thank my family and friends for their constant encouragement and support throughout my studies. Their confidence in me has been a source of motivation during both the rewarding and challenging moments of this journey.

Declaration on the Use of AI-Assisted Technologies

In preparing this dissertation, the author used generative AI tools (Gemini and Claude) for limited technical assistance only: support with coding and debugging (Cast3M scripts and Python post-processing), language editing and structure suggestions, and help in quickly organising and synthesising data during the research process. The scientific content, the analyses, the interpretation of the results and the conclusions presented in this work were produced by the author.

No AI tool was used to generate scientific conclusions, to fabricate or alter data, or to replace the author's own critical analysis. All AI-assisted material was reviewed, verified and corrected where necessary, and integrated under the author's responsibility. The author retains full responsibility for the content of this dissertation.

Abstract

Concrete spalling, the sudden explosive detachment of concrete surface when concrete is exposed to rapidly rising temperature, is a major threat to the integrity of structures subjected to fire or to severe accident conditions, such as nuclear containment buildings. It results from coupled thermo-hydro-mechanical (THM) process of the material, which is still incompletely understood and remains difficult to predict.

The work studies the THM behaviors of concrete under severe accident conditions using a stochastic poro-mechanical approach. In a coupled THM model, the thermo-hydric subsystem is solved monolithically, and the mechanical problem (linear elasticity with a MAZARS damage law) is coupled in a staggered manner. While the full THM phenomenon is explored theoretically, the numerical implementation in CAST3M and validation performed in this work are specifically focused on the Thermo-Hydro-Chemical (THC) scope.

Unlike the traditional approach of TH coupling in [6], this work adopts the new THC approach of [14]. The transport and storage properties are redefined using the *hydration degree*, this ensures that the initial hydro-chemical state of the material is integrated into the model.

The model is then calibrated and validated against the experiment conducted by Kalifa [11], which served as a reference benchmark, with the aim of reproducing the measured temperature and gas pressure evolutions within concrete specimens. The temperature field is reproduced accurately, whereas the gas pressure cannot be matched with the same confidence. This raises the question of whether the gauges truly recorded the pore gas pressure.

Finally, the outcomes of the calibrated model contribute as the input for a stochastic assessment of the model, carried out with OPENTURNS. Therefore, it provides insights of the correlation and sensitivity analyses that identify the dominant parameters; a reliability analysis quantifies the probability of failure and the risk of spalling; and a random-field description of material heterogeneity is shown to produce more critical predictions than the homogeneous assumption.

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Chapter 1

Introduction

1.1 Context and problem statement

Concrete structures exposed to extreme heat undergo a complex interpaly of thermal, hyric, and mechanical phenomena that con lead to progressive degradation, and in most severe cases, to *spalling*, the sudden, explosive detachment of concrete layers from the exposed surface.

This phenomenon has been a critical concern for civil infrastructure such as nuclear containment buildings and tunnels, where the consequences of spalling as the loss of cover and cross-section can damage the load-bearing capacity and functions of the structure.

Recently, its catastrophic consequences are also relevant in aerospace engineering. In 2023, a SpaceX rocket test has failed due to the destructive power of concrete spalling [13]. As the rocket ignited on its concrete launch pad, the immense heat generated triggered explosive spalling. The debris seen flying around the launch site consisted of sharp concrete flakes. These flakes shot upwards, severely damaged the test field and the rocket itself, ultimately causing the rocket test launch to fail.

(a) The crater in the launchpad after SpaceX's Starship took off.

(b) Debris litters the Starship launchpad, with damaged fuel tanks visible in the background.

Figure 1.1: Catastrophic consequences of explosive concrete spalling during the SpaceX Starship test flight.

1.1.1 Concrete behaviour under high temperature

When concrete is heated rapidly, several coupled physical processes occur simultaneously

- **Thermal:** Steep temperature gradients develop throughout the concrete's cross-section, causing differential thermal expansion and generating compressive stresses at the heated surface.
- **Hydric:** Both free and chemically bound water evaporate and move through the pore network. In low-permeability concretes such as high-performance concrete (HPC), vapor cannot escape fast enough, resulting in pore pressure build-up behind a quasi-saturated *moisture clog*.
- **Mechanical:** The combined effects of thermal stresses and pore pressure build-up on the solid skeleton can exceed the tensile strength, which is degraded by temperature. This initiate cracks parallel to the heated surface and potentially lead to explosive spalling.

These three fields are not independent: they interact through a network of coupling mechanisms that any predictive model must reproduce. Despite extensive experimental work, the mechanisms governing spalling are still incompletely understood and no single, universally accepted theory has emerged [6].

Because neither a purely thermal nor a purely hydric analysis can capture this full picture, a fully coupled THM model is strictly required to understand the material's response and assess structural safety. The detailed description of these THM behaviors and their complex interactions is deferred to Appendix A.

1.1.2 The initial hydro-chemical state of concrete

The response of concrete to fire does not depend only on the heat: it also depends strongly on the state of the material *before* heating. The calcium–silicate–hydrates (C–S–H) formed during cement hydration control the strength, the porosity and the amount of chemically bound water that can later be released by dehydration.

The degree to which hydration has progressed, together with the residual moisture content inherited from curing and drying, therefore conditions the transport and storage properties that govern pore pressure build-up. Accounting for this initial hydro-chemical state is the key idea behind recent thermo-hygro-chemical (THC) models [14], and it motivates the formulation adopted in this work.

1.1.3 Limitations of existing models

Existing THM models present several limitations that this work seeks to address:

- The initial hydro-chemical state is often treated implicitly, through empirical functions of temperature only, rather than through an explicit hydration-degree description;
- The uncertainty in the material parameters is rarely assessed: properties are typically taken as homogeneous, deterministic values, so the influence of their variability on the predicted response is left unquantified.

1.2 Objectives and scope of the report

This work aims to numerically investigate the behaviour of high-performance concrete exposed to high temperatures. While the full Thermo-Hydro-Mechanical (THM) phenomenon is explored theoretically to understand the root causes of spalling, the actual numerical implementation and validation performed in this internship are strictly limited to the Thermo-Hydro-Chemical (THC) scope. This limitation is due to the timeframe of the internship and the fact that the reference experimental data (Kalifa test) only provide temperature and gas pressure measurements, lacking mechanical counterparts (e.g., strains, damage) for validation.

The remainder of the report is structured as follows:

- Chapter 2 presents the theoretical framework of concrete at high temperature. It outlines the full THM coupling mechanisms conceptually, but details specifically the governing equations of the THC model, incorporating the hydration-degree formulation following [14].

- Chapter 3 describes the reference experiment of [11] and its numerical setup in Cast3M using the THC model.
- Chapter 4 reports the calibration of the temperature and gas-pressure responses, validates the model, and discusses the limits of the gas-pressure measurements.
- Chapter 5 performs a stochastic assessment of the calibrated THC model with OpenTURNS (Sobol sensitivity analysis, reliability analysis, and spatial heterogeneity).
- Chapter 6 summarizes the main findings and outlines perspectives for future fully-coupled THM developments.

1.3 Research environment and Computational tools

This work was carried out at the *Laboratoire Sols, Solides, Structures – Risques* (3SR), Université Grenoble Alpes, which provided the scientific environment and the expertise in multiphysics modelling and risk analysis.

The coupled THM model is implemented in Cast3M (or Castem), an open-source finite-element code developed at CEA. Compared with closed multiphysics platforms, COMSOL Multiphysics, Cast3M offers greater transparency and control for implemnting tightly coupled THM, while COMSOL offers a more user-friendly GUI.

The probabilistic uncertainty quantification relies primarily on OpenTURNS, an open-source library based on Python used to generate the random samples, propagate them through the model, and compute the Sobol indices and reliability. Alternatively, uncertainty quantification can also be operated directly using Cast3M; while Cast3M is less powerful regarding advanced sensitivity and reliability analyses, it provides the essential capability of representing spatial material heterogeneity using random fields Section 5.7.2.

Finally, Python (in Visual Studio Code) is also used to automate the calibration workflow and post-process the results, and LaTeX to typeset this report.

Chapter 2

Theoretical framework

[REVIEW] The old content of this chapter is mostly moved to Appendix A. This chapter is a draft version and its outline is as follow.

2.1 Thermo-hydro-mechanical behavior and spalling mechanisms

When a concrete member is exposed to a rising temperature, a sequence of strongly interacting thermal, hydric, and mechanical processes develops through the thickness. Heat conduction triggers phase changes (evaporation, dehydration), generating vapour that is transported and recondenses to form a quasi-saturated moisture clog. Concurrently, thermal gradients and pore pressure build-up induce mechanical stresses. The full mathematical derivation of these THM behaviors is detailed in Appendix A.

These interacting phenomena are the root cause of *spalling*. Explosive spalling generally results from the *combined action* of pore pressure build-up and thermal stress (Figure 2.1).

The spalling condition can be written schematically as:

$$\sigma_{thermal}(T) + \sigma_{pore}(p_g, p_c, S_l) \geq f_t(T), \tag{2.1}$$

where $\sigma_{thermal}$ is the thermal stress from differential dilation, σ_{pore} the stress from pore-pressure loading, and $f_t(T)$ the temperature-degraded tensile strength. A comprehensive breakdown of this combined spalling mechanism is provided in Appendix A.6 of Appendix A.

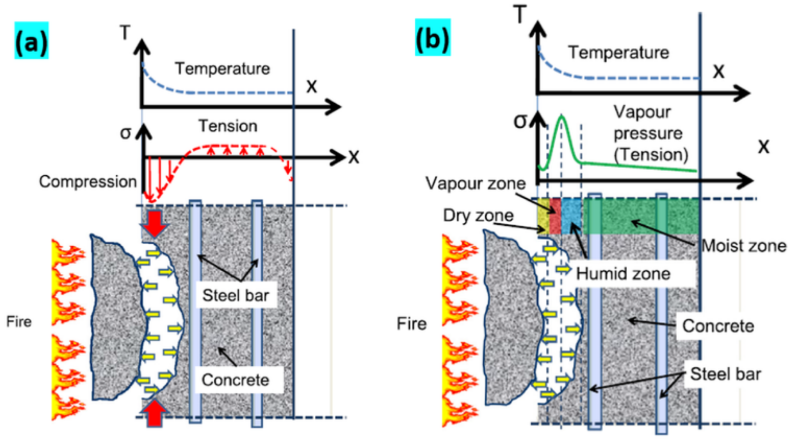


Figure 2.1: Spalling combined mechanisms [5]

2.2 Improved thermo-hydro-chemical (THC) model

2.2.1 Motivation

In the TH model above, dehydration enters only as a prescribed source term $\dot{m}_{dehyd}(T)$ and the porosity follows a simple linear law $\phi = \phi_0 + A_\phi(T - T_0)$ (Eq. **??** in Appendix **??**). This treatment ignores three facts that strongly influence spalling: (i) the *chemical state* of the cement paste before the accident depends on its hydration/curing history; (ii) the release of chemically bound water from the C–S–H is the chemical process that controls porosity, permeability and strength at high temperature; and (iii) chemical damage is essentially *irreversible*. The improved model adopted here, following Sciumè, Moreira and Dal Pont [14], resolves the chemical field explicitly and couples it to the thermal and hydric fields.

2.2.2 Chemical state variables

Unlike standard TH models, the THC framework explicitly resolves the chemical state of the cement paste. The evolution of the microstructure depends on two main processes [14]:

- **Hydration degree $\Gamma(t) \in [1]$:** Defines the formation of Calcium-Silicate-Hydrates (C-S-H) during curing and aging. Its evolution is governed by an Arrhenius-type chemo-kinetic law (detailed in Appendix A.3.2), driven by time, temperature, and relative humidity.
- **Dehydration degree $F(T) \in [1]$:** Represents the irreversible release of chemically bound water at high temperatures (typically $T > 105°\text{C}$).

To couple these effects, the **effective hydration degree** $\tilde{\Gamma}$ is introduced as the primary internal variable describing the state of the solid matrix:

$$\tilde{\Gamma} = (1 - F(T))\,\Gamma \tag{2.2}$$

The transport and mechanical properties, such as porosity $\phi(\tilde{\Gamma})$ and intrinsic permeability $K(\tilde{\Gamma})$, are explicitly updated as functions of this effective hydration state. The specific evolution laws for these microstructural properties are detailed in Appendix A.3 of Appendix A.

2.2.3 Resulting coupled system

The complex mathematical derivation of the governing equations, based on Powers' stoichiometry and the averaging theory for multiphase porous media, is detailed in Appendix A. By incorporating the chemical couplings, the final system reduces to three highly non-linear conservation equations for the state variables (p_g, p_c, T):

1. Dry-air conservation:

$$\frac{\partial}{\partial t}\big(\phi(1 - S_l)\rho_a\big) + \nabla \cdot (\mathbf{J}_a) = (1 - S_l)\rho_a \left(a_\phi \Omega \frac{\partial \tilde{\Gamma}}{\partial t} \right) \tag{2.3}$$

2. Total water-species conservation:

$$\frac{\partial}{\partial t}\big(\phi S_l \rho_l + \phi(1 - S_l)\rho_v\big) + \nabla \cdot (\mathbf{J}_l + \mathbf{J}_v) = [S_l \rho_l + (1 - S_l)\rho_v] \left(a_\phi \Omega \frac{\partial \tilde{\Gamma}}{\partial t} \right) - 0.228 c \xi_\infty \frac{\partial \tilde{\Gamma}}{\partial t} \tag{2.4}$$

3. Energy conservation:

$$\overline{\rho C_p} \frac{\partial T}{\partial t} + \mathbf{J}^{adv}_{energy} - \nabla \cdot (\lambda \nabla T) = -H_{vap} \dot{m}_{vap} + (H_{vap} 0.228 c \xi_\infty + L_{hyd}) \frac{\partial \tilde{\Gamma}}{\partial t} \tag{2.5}$$

where a_ϕ is the porosity evolution parameter, c is the cement content, and ξ_∞ is the maximum hydration fraction. The explicit formulation of these stoichiometric constants (e.g., the bound water $0.228 c \xi_\infty$ and the latent heat L_{hyd}) as well as the thermal activation energy E_a/R governing $\tilde{\Gamma}$ are detailed in Appendix A.3.1 and Appendix A.3.2 of Appendix A.

2.3 Coupling to the mechanical field

The mechanical response is driven by the converged TH/THC fields through a *one-way* (staggered) coupling: at each time step the fields (T, p_g, p_c) enter the mechanical problem as prescribed loading, and the resulting damage $\mathcal{D}$ feeds back weakly to the transport properties via permeability evolution at the next step. The TH/THC fields act through two routes:

- **Free strains.** The total strain is decomposed as $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_\sigma + \boldsymbol{\varepsilon}_{th} + \boldsymbol{\varepsilon}_{sh}$, where $\boldsymbol{\varepsilon}_{th}$ is the thermal dilation (driven by T) and $\boldsymbol{\varepsilon}_{sh}$ the capillary shrinkage (driven by p_c). Creep is neglected, so $\boldsymbol{\varepsilon}_\sigma = \boldsymbol{\varepsilon}_{el}$.
- **Pore-pressure loading.** Through Bishop's effective stress $\boldsymbol{\sigma'} = \boldsymbol{\sigma}^{Total} + \alpha\, p_s\, \mathbf{I}$, the average pore pressure $p_s = p_g - S_l p_c$ enters the load vector as an external loading; its build-up shifts $\boldsymbol{\sigma'}$ towards tension and promotes cracking when $f_t(T)$ is exceeded.

Together with the spalling criterion (2.1), this closes the physical loop between the transport and mechanical fields. The mechanical constitutive model, its fracture-energy regularisation, and the spatial/temporal discretisation of the coupled problem are reported in full in Appendix A.

Note on the scope of the study: Due to the limited timeframe of the internship and the fact that the experimental data used for validation (Kalifa test) solely provide temperature and gas pressure measurements, the mechanical module cannot be rigorously validated. Therefore, while this staggered mechanical coupling is mathematically formulated and designed as an easily pluggable module, the actual numerical simulations performed in the subsequent chapters are restricted to the TH(C) model.

Chapter 3

Experimental Reference and Numerical Model

[REVIEW] This chapter is a draft version and its outline is as follow.

3.1 The Kalifa experiment

Purpose. The campaign of Kalifa et al. [11] quantifies the *thermo-hydral* process responsible for spalling. When concrete is heated on one face, free and chemically bound water are released as vapour; part migrates inward, recondenses in cooler regions and forms a quasi-saturated *moisture clog* that blocks gas evacuation and triggers a sharp pore-pressure build-up behind the drying front. The test measures *simultaneously and continuously* the temperature, the gas (pore) pressure at several depths, and the global mass loss, providing a quantitative dataset for validating coupled codes.

Set-up and instrumentation. A prismatic specimen of $30 \times 30 \times 12$ cm^3 is heated on one 30×30 cm^2 face by a radiant heater (up to 5 kW, 800 °C) placed 3 cm above the surface. The lateral faces are insulated with porous ceramic blocks, so heat and mass transfer are essentially one-dimensional. The specimen rests on a balance recording its mass. Five gauges measure pressure and temperature at 10 mm, 20 mm, 30 mm, 40 mm and 50 mm from the heated face, and a sixth tube measures the surface temperature at 2 mm depth.

Concrete mix. Only the high-performance concrete (HPC, mix M100-C) is retained for the modelling: it exhibits the most pronounced thermo-hydral response (measured gas-pressure peak ≈ 40 bar) and an initial liquid saturation $S_{l,0} \approx 0.77$. Its mixture proportions, after Kalifa et al. [11], are given in Table 3.1: a CEM I 52.5 cement blended with

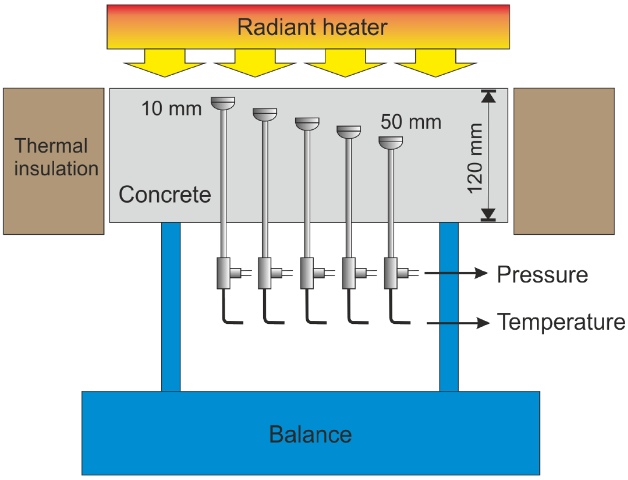


Figure 3.1: Kalifa experiment setup [6]

silica fume at a low water/binder ratio, giving the high compactness and low permeability responsible for the strong pressurisation. The lumped quantities fed to Powers' model (c, aggregate, w/c, s/c) are those listed in Table 3.2.

Table 3.1: Mixture proportions and strength of the HPC (M100-C) modelled in this work, after Kalifa et al. [11].

Constituent	M100-C [$\mathrm{kg\,m^{-3}}$]
Cement CPA, CEM I 52.5	377
Silica fume	37.8
Water	124
Superplasticiser (Résine GT)	12.5
Retarder (Chrytard)	2.6
Sand, silico-calcareous (0/4 mm)	432
Sand, calcareous (0/5 mm)	439
Aggregate, calcareous (5/12.5 mm)	488
Aggregate, calcareous (12.5/20 mm)	561
Water/binder ratio	0.34
Nominal strength [MPa]	91.8 ± 0.8
Initial saturation $S_{l,0}$	≈ 0.77

Heating scenario and measured data. In the experiment the specimen is heated on one face by a radiant heater applied as a step signal, the heater temperature being set to one of 450 °C, 600 °C and 800 °C. The 600 °C test is retained as the reference loading, since it develops a complete pressurisation/relief cycle while preserving the quasi-1-D transfer condition. Under this loading the typical response combines a temperature rise with a short plateau near 200 °C (intense vaporisation), a two-stage gas-pressure history (pressurisation, then relief as the drying front passes the gauge), and a continuous mass loss; in HPC the pore pressure exceeds the saturating vapour pressure $p_{vs}(T)$.

These three quantities — temperature and gas pressure at the gauge depths and the global mass loss — are the validation targets of this work. They are not plotted here: to keep the experimental curves together with the simulated response, they are reported alongside the numerical results in the calibration in Section 4.1).

3.2 Numerical model

Scope of the numerical simulation (THC model). It is important to note that while the full Thermo-Hydro-Mechanical (THM) theoretical framework is derived in Chapter 2 and Appendix A, the numerical application in this chapter is restricted to the Thermo-Hydro-Chemical (THC) model. Because the reference Kalifa experiment only provides temperature and gas pressure data, there are no experimental mechanical counterparts (e.g., strain or damage measurements) available to validate the structural response. Consequently, the staggered mechanical module—though formulated as an easily pluggable component—is kept out of the current internship scope.

Geometry and mesh. The test is reproduced by a two-dimensional plane model of the 120×120 mm^2 cross-section (Figure 3.2), with the heated (hot) face at the top and the cold face at the bottom; the two lateral faces are thermally and hydrically insulated, so the model reproduces the quasi-1-D transfer of the experiment. The domain is discretised with 24×24 four-noded quadrilateral (QUA4) elements (the reference paper used 28×28 [6]). Field histories are extracted along the central vertical line ($x = 60$ mm) at depths of $0, 10, 20, 30, 40, 50$ and 120 mm from the hot face, matching the gauge positions. A graded mesh refined toward the hot face (~ 1 mm) is also available for the mesh-sensitivity study.

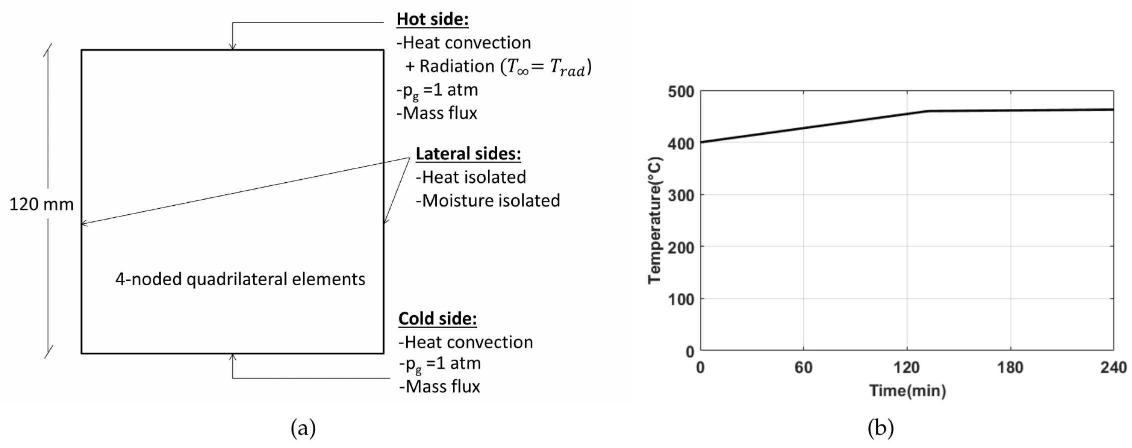


Figure 3.2: (a) Model set-up and boundary conditions and (b) radiator temperature used in the modelling, after Dauti [6].

Material parameters. The reference parameters corresponding to the HPC M100 mix are summarised in Table 3.2. Instead of treating high-temperature concrete as a purely empirical material, the inputs feed directly into the THC constitutive laws:

- **Hydration kinetics ($A_i, A_p, \Gamma_p, \zeta, E_a/R, L_{hyd}$):** These parameters define the normalized chemical affinity $\mathcal{A}(\Gamma)$ and the thermal activation E_a/R (input as EAR) which drive the hydration rate during the curing phase. The latent heat L_{hyd} (input as Q_{lat}) defines the thermal energy released. The full physical meanings of these terms are detailed in Appendix A.3.2.
- **Permeability evolution (K_0, A_Γ, B_K):** Rather than applying an empirical temperature-dependent multiplier, the intrinsic permeability K is explicitly coupled to the effective hydration degree $\tilde{\Gamma}$ via the relation:

  $$K(\tilde{\Gamma}) = K_0 \cdot 10^{A_\Gamma (1-\tilde{\Gamma})^{B_K}} \tag{3.1}$$

  where K_0 is the initial intrinsic permeability (often denoted K_1 in the numerical code). A_Γ dictates the magnitude of permeability increase as C-S-H decomposes, while B_K controls the curve shape. Further theoretical derivations are provided in Appendix A.3.4.

The porosity, the water content at full hydration, and the initial hydration field are internally computed using Powers' stoichiometric model (Appendix A.3.1 in Appendix A).

Thermal input and boundary conditions. The boundary set-up follows the modelling of Dauti [6] (Figure 3.2). On the hot face a combined convection–radiation condition is imposed,

$$-\lambda \frac{\partial T}{\partial n}\bigg|_{\text{hot}} = h^{\text{hot}}(T_\infty - T) + \epsilon\sigma\big(T_\infty^4 - T^4\big), \tag{3.2}$$

with $h^{\text{hot}} = 8\,\mathrm{W\,m^{-2}\,K^{-1}}$, emissivity $\epsilon = 0.7$, and a prescribed radiator temperature $T_\infty(t)$ that rises to $\approx 400\,°\mathrm{C}$ at the onset of the accident and then increases quasi-linearly to $\approx 466\,°\mathrm{C}$ over the ≈ 240 min accident (Figure 3.2b); the radiator is switched off afterwards. The cold face exchanges heat by convection only ($h^{\text{cold}} = 8.3\,\mathrm{W\,m^{-2}\,K^{-1}}$, $T_\infty = 20\,°\mathrm{C}$). On both faces the gas pressure is fixed at atmospheric ($p_g = p_{atm}$) and a convective vapour flux $q_w = h_g(\rho_v^s - \rho_v^\infty)$ is applied.

The mass-transfer coefficient h_g increasing from 0 (sealed) to 2×10^{-3} (drying) and $1.8 \times 10^{-2}\,\mathrm{m\,s^{-1}}$ (accident). It is important to distinguish these transfer coefficients: while the convective heat transfer coefficients ($h^{\text{hot}}, h^{\text{cold}}$) govern the exchange of thermal energy across the boundary, the mass-transfer coefficient (h_g) specifically dictates the rate at which water vapour escapes from the porous network to the surrounding environment.

The radiator history is adapted from Dauti [6]: rather than a raw furnace curve, it is shaped so that the resulting *outer-surface* temperature of the specimen reproduces a

Table 3.2: Reference parameters of the THC model (HPC M100).

Parameter	Symbol	Value
Mix		
Cement content	c	$377\,\mathrm{kg\,m^{-3}}$
Aggregate content	a	$1920\,\mathrm{kg\,m^{-3}}$
Water/cement ratio	w/c	0.34
Silica/cement ratio	s/c	0.1
Solid density	ρ_s	$2625.8\,\mathrm{kg\,m^{-3}}$
Hydration / affinity		
Affinity parameters	A_i, A_p	6, 1250
	Γ_p, ζ	0.24, 48
Activation	E_a/R (EAR)	5369 K
Hydration heat	L_{hyd} (Q_{lat})	$114.3\,\mathrm{MJ\,m^{-3}}$
Thermal		
Solid heat capacity	C_{ps} (CS)	$948\,\mathrm{J\,kg^{-1}\,K^{-1}}$
Solid conductivity	λ_{d0} (KS)	$1.67\,\mathrm{W\,m^{-1}\,K^{-1}}$
Transport / retention		
Reference permeability	K_0 (K_1)	$2 \times 10^{-19}\,\mathrm{m^2}$
Permeability coeff.	A_Γ, B_K	3, 1
Liquid permeability	K_{li}	20 000
Desorption curve	a_1	18.6 MPa
	b_1, c_1	2.27, 1.5

physically reasonable response; the thermal boundary and bulk conduction parameters (emissivity, convective heat-transfer coefficient, solid thermal conductivity, and solid heat capacity) are subsequently re-calibrated against the temperature measurements in Chapter 4 (Section 4.2).

3.3 Application of the THC model

3.3.1 Implementation in Cast3M

The THC model is implemented and solved within the open-source finite element code Cast3M. The coupled convective mass and heat boundary conditions are strongly non-linear: the fluid densities entering the convective fluxes must be expressed in terms of the state variables (p_c, p_g, T). Because this does not admit a closed algebraic form suitable for standard discretisation, the implementation relies on a custom pre-release build of Cast3M (developed by G. Sciumè, S. Dal Pont, and D. Dauti), which provides the dedicated operators required to resolve the highly coupled THC formulation efficiently.

3.3.2 Solution strategy and numerical robustness

At high temperature, the THC system is *strongly nonlinear*. The Newton iterations of the nonlinear `PASAPAS` integrator may therefore *diverge*. To restore convergence and ensure numerical stability, the following measures are implemented:

- **Increasing the relaxation parameter** θ toward a fully implicit time integration ($\theta = 1$, used here). While a fully implicit scheme slightly reduces the temporal accuracy compared to the Crank-Nicolson method ($\theta = 0.5$), it efficiently damps the oscillations of the nonlinear coefficients and strongly favors convergence. The mathematical formulation of this θ-method is detailed in Appendix A.10.2.

- **Reducing the time step** Δt: To compensate for the loss of accuracy caused by setting $\theta = 1$, and to further enlarge the convergence radius, Δt is drastically reduced: from 600 s during aging down to 5 s and 15 s across the thermal ramp. Furthermore, Δt is strictly reduced to 1 s during the first and last minutes of the thermal ramp where temperature gradients are extremely high. This simultaneous reduction ensures both numerical accuracy and optimal convergence (see calibration in Section 4.2).

- **Retaining linear elements (`QUA4` over `QUA8`):** In standard finite element modeling, upgrading to higher-order elements (such as 8-node quadrilaterals) is often a tempting strategy to improve spatial accuracy. However, in highly non-linear multiphase flows, these higher-order elements are highly susceptible to spatial numerical oscillations, which can unphysically drive the gas pressure to negative values and cause the computation to crash. Following expert discussions with Prof. Stefano Dal Pont during the troubleshooting phase, the default 4-node linear quadrilateral elements (`QUA4`) from the original sample code were deliberately retained. `QUA4` elements inherently suppress these numerical artifacts, ensuring a physically valid and strictly positive pressure field throughout the severe thermal gradients.

- **Mesh density considerations:** Although localized mesh refinement toward the hot face (e.g., a graded mesh down to ~ 1 mm) is theoretically known to limit inter-element jumps of variables and improve stability, it was ultimately *not applied* in the final actual code. The mesh was deliberately kept at a regular, unrefined 24×24 grid. Simplifying the spatial discretization and falling back to the original stable mesh was a necessary step to isolate physical instabilities from meshing-induced errors, ensuring the model runs successfully until the end.

- **Imposing an artificially low initial relative humidity (RH_0):** For the early-age (sealed) curing phase, setting a physically realistic initial relative humidity (e.g., $\mathrm{RH}_0 = 0.98$) causes the non-linear solver to diverge immediately due to severe capillary pressure gradients. To bypass this, RH_0 is deliberately lowered to 0.5. While physically unrealistic for a fresh concrete specimen, this numerical relaxation is strictly required to initialize the computation and, importantly, it does not negatively impact the final gas-pressure calibration during the high-temperature accident phase.

- **Tuning the thermal radiation**: The thermal radiation input (emissivity) at the heated boundary is deliberately adjusted and reduced to prevent excessive surface overheating. $\epsilon = 0.85$ in [6] is reduced to $\epsilon = 0.7$. This keeps the resulting temperatures within a tractable range.

- **Balancing permeability parameters (K_0 and A_Γ):** The coefficient A_Γ drastically affects the permeability evolution. A high value triggers severe pressure gradients and numerical singularities. As will be detailed in the gas pressure calibration (Section 4.3.1), to ensure initial numerical stability, the starting point was deliberately set with a relatively high initial permeability $K_0 = 2 \times 10^{-19}$ m^2 and a lower coefficient $A_\Gamma = 3$.

All these robustness measures and numerical stabilization strategies were thoroughly investigated and implemented by the author, requiring a substantial investment of time and effort to successfully debug and execute the highly non-linear THC simulations without divergence.

3.3.3 Simulation timeline and phases

A key feature of the THC application is the chain of chemo-physical effects: the hygrochemical state preceding the accident is not prescribed arbitrarily but is obtained from preliminary simulations. The curing and aging history fixes the hydration state $\tilde{\Gamma}$, which in turn controls the porosity, the saturation curve, and the intrinsic permeability. The full simulated history is structured into three consecutive phases, executed via staged `PASAPAS` computations in Cast3M:

1. **Early age (0–3 days):** The concrete is assumed to be sealed and hydrating under adiabatic conditions. The computation starts from $T_0 = 20\,°\mathrm{C}$, $p_{g,0} = 101\,325\,\mathrm{Pa}$, $\Gamma_0 = 0$, and an artificially lowered relative humidity $\mathrm{RH}_0 = 0.5$ (see Section 3.3.2 for the numerical justification of this value).

2. **Aging and drying (3 days–1 year):** The specimen is exposed to heat and moisture convection, with the external humidity progressively reduced down to 0.50. This stage establishes the self-consistent profiles of hydration degree, liquid saturation S_l, and internal relative humidity just before the fire event.

3. **Accident phase (0–240 mins):** The heating ramp is applied to the hot face (convection and radiation). The temperature rises sharply, triggering the dehydration

of C-S-H and the subsequent pore-pressure build-up. This phase represents the core of the TH analysis.

Cooling phase omission: Although the Kalifa experimental campaign continued to record temperature and gas-pressure data during the cooling phase (from 240 up to 600 minutes), this final phase is *not* modeled in the present work. Since the primary objective of this study is to investigate the mechanisms triggering explosive spalling—which intrinsically occurs during the aggressive heating and peak-pressure phase—the numerical simulation is deliberately stopped at $t = 240\,\text{min}$. Consequently, the experimental cooling data is excluded from the comparative plots in the calibration phase (Chapter 4).

Chapter 4

Model Calibration and Validation

4.1 Model calibration

The accuracy of TH/THC model relies on a large number of material parameters – intrinsic permeability, thermal conductivity and heat capacity, sorption properties, surface exchange coefficients and a complex set of constitutive laws describing their evolution with temperature and dehydration.

Many critical properties at high-temperature such as the evolution of intrinsic permeability, sorption isotherms are extremely difficult or even impossible to measure directly through experimental techniques and the values often contain uncertainties due to the intrusive nature of the sensors. Furthermore these are not measured directly in the Kalifa campaign.

A pre-determined parameter set therefore cannot be expected to reproduce the experiment. Consequently, a calibration procedure for the numerical models is strictly required to identify, within physically admissible ranges, the parameter set that makes the model reproduce the measured temperature and gas-pressure histories. Only once this reference case is calibrated and validated can the model be trusted for uncertainty studies in Chapter 5.

4.1.1 Calibration strategy

At the first approximation, the thermal problem is weakly coupled to the hydric one: the temperature field drives evaporation, dehydration and the gas-pressure build-up, while the effect of the hydric state on the temperature field is relatively small within the experimental range. This hierachy promotes a two-stage calibration process:

1. **Stage 1 – Temperature.** The temperature evolution is calibrated independently

of the pressure field. In this stage, the *surface* (boundary) parameters should be calibrated first, then the *bulk* conduction parameters.

2. **Stage 2 – Gas pressure.** With the reliably calibrated temperature field as a fixed thermal loading, the hydric parameters gorvening the gas-pressure evolution are adjusted, from the most influential to the secondary ones.

The calibration targets are the temperature and gas-pressure histories measured at 10, 20, 30, 40 and 50 mm from the heated face, together with the near-surface temperature recorded at 2 mm (denoted the 0 mm measurement).

4.1.2 Automated parameter sweep

The sweeps are driven by a Python script that launches Cast3M repeatedly, one run per parameter combination, without manual intervention. For each stage the parameter values to be tried are listed explicitly, and `itertools.product` builds every possible combination of those lists; a single loop then feeds each combination to Cast3M in turn.

This automation relies on two supporting pieces of code.

- First, the Cast3M input code itself exports the results of each run to a `.csv` file – the temperature and gas-pressure histories at the measurement points.
- Second, a separate Python post-processing script reads these `.csv` files and plots the computed histories against the experimental measurements [11] and also against the results from Dorjan Dauti's thesis [6]. All combinations then can be visually compared, and the best-fit set is selected by the author based on this comparison.

For example, in the gas-pressure calibration stage, the two parameters swept are the intrinsic permeability K_0 (denoted `KINT`) and the permeability–hydration coefficient A_Γ (denoted `AK`), with the trial values:

- $K_0 \in \{1{\times}10^{-19},\ 2{\times}10^{-19},\ 1{\times}10^{-20},\ 5{\times}10^{-20},\ 7.5{\times}10^{-20},\ 8.5{\times}10^{-20}\}$ m^2 (6 values);
- $A_\Gamma \in \{3,\ 4,\ 5,\ 6,\ 7,\ 8\}$ (6 values).

These two lists give $6 \times 6 = 36$ parameter pairs, i.e. 36 Cast3M runs. The same driver is reused by simply changing the value lists.

In every stage, a wider range of each parameter is first explored to bracket the admissible values; the search is then narrowed to the few representative values reported in the following sections, which are the ones shown in the figures.

[REVIEW] The following sections are currently under review and need to be rewritten

4.2 Temperature calibration

4.2.1 Surface temperature (00mm): boundary heat exchange coefficient

The first quantity calibrated is the temperature of the heated face. In the experiment, heat is supplied by a radiant heater on the top-hot face, so the surface energy balance is mostly influneced by radiation and convection. See Equation (A.2) and Equation (A.4):

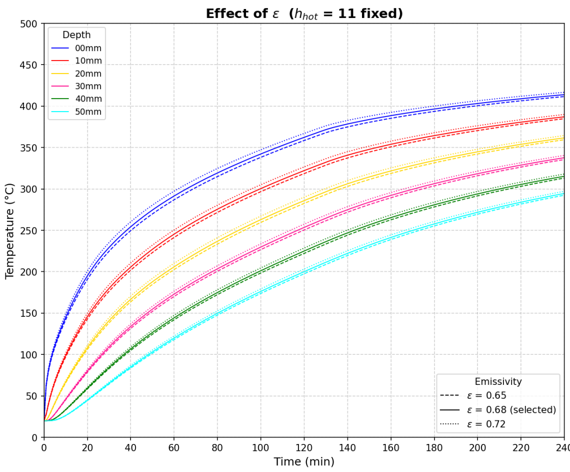
$$q_{00\,\mathrm{mm}} = h^{\mathrm{hot}}\,(T_\infty - T_s) + \varepsilon\,\sigma\,\left(T_{\mathrm{sur}}^4 - T_s^4\right), \tag{4.1}$$

where h^{hot} is the convective heat-transfer coefficient at the top-hot surface, ε the surface emissivity (radiative coefficient), σ the Stefan–Boltzmann constant, T_{sur} and T_∞ are the surrounding temperature provided by the radiant heat source and T_s the surface temperature.

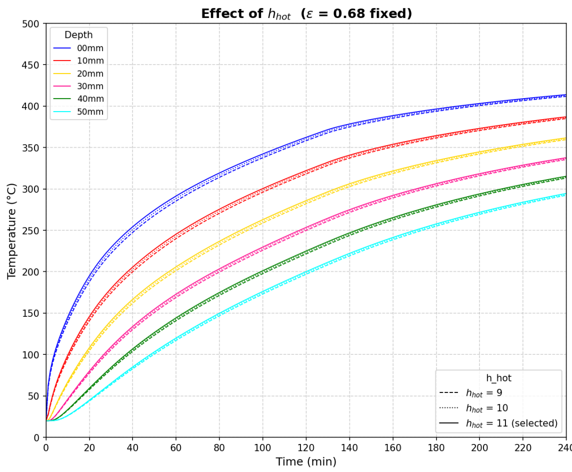
The two boundary coefficients h^{hot} and ε are adjusted so that the computed 0 mm history matches the measured near-surface (2 mm) response. At this point the conduction parameters are held at their reference values; only the energy input at the surface is tuned.

The surface response is calibrated over three values of ε (0.65, 0.68, 0.72) and three of h_c (9, 10, 11 $\mathrm{W\,m^{-2}K^{-1}}$).

Because the resulting curves differ only slightly, the full 3×3 grid is not informative; instead the effect of each parameter is shown one at a time, varying it while the other is held at its retained value (Figures 4.1a and 4.1b). The pair retained at this stage is $\varepsilon = 0.68$ and $h^{\mathrm{hot}} = 11\ \mathrm{W\,m^{-2}K^{-1}}$.



(a) Sensitivity to the emissivity ε ($h_c = 11$ fixed).



(b) Sensitivity to the convective coefficient h_c ($\varepsilon = 0.68$ fixed).

Figure 4.1: Surface (0 mm) temperature calibration, compared with the measured near-surface response.

Two distinct effects are concluded from Figures 4.1a and 4.1b:

- The emissivity ε is the most sensitive parameter (Figure 4.1a): because the radiative term acts in a strong non-linear way as T^4 in Equation (4.1), a small increase in ε raises the surface temperature substantially, and mostly in the high-temperature part of the curve, where radiation dominates the heat input.
- The convective coefficient h^{hot} acts more gently and mainly on the *early* part of the curve (Figure 4.1b): increasing h^{hot} raises the initial heating rate (the linear $h^{\text{hot}}(T_\infty - T_s)$ term is largest while T_s is still low), with only a slight additional rise in the tail.

In brief, ε sets the level of the high-temperature plateau while h^{hot} controls the initial ramp; tuning the two together reproduces the measured surface temperature evolution.

4.2.2 In-depth temperature: thermal conductivity and heat capacity

Once the surface temperature is reproduced, the temperature rise at depth (10–50 mm) is governed by heat conduction through the material, i.e. by the volumetric heat capacity ρC_p (**??**) and the thermal conductivity λ (Equation (A.9) and Equation (A.10)), both given in Appendix A. Most of the quantities in these laws (ϕ, S_l, the phase densities ρ and the temperature T) are imposed by the state of the material. The two genuinely adjustable inputs are:

- the solid specific heat C_{ps} (denoted `CS` in the model), which enters the heat capacity through Eq. **??**. Although a temperature dependence is defined in Appendix A, in the present implementation it is kept *constant* – it is not updated with temperature;
- the reference dry conductivity λ_{d0} (denoted `KS` in the model), which scales the whole conductivity law.

A subtlety arises here: adjusting the surface condition in Section 4.2.1 changes the energy that actually enters the specimen, which in turn shifts the in-depth histories. Consequently λ_{d0} and C_{ps} cannot be calibrated in isolation – they must be *re-calibrated* jointly with the surface condition.

The calibration therefore proceeds as a short journey, starting from the parameter set carried over from the surface stage and adjusting one quantity at a time.

Starting point. The run is first launched with the reference set inherited from the surface calibration in Section 4.2.1: $\varepsilon = 0.68$, $h^{\text{hot}} = 11\ \mathrm{W\,m^{-2}K^{-1}}$, $C_{ps} = 948\ \mathrm{J\,kg^{-1}K^{-1}}$ and $\lambda_{d0} = 1.67\ \mathrm{W\,m^{-1}K^{-1}}$. Compared with the experiment and with the earlier results of Dauti (Figure 4.2), the temperature is already well reproduced near the surface, but the deeper probes (30, 40 and 50 mm) are over-predicted: the in-depth temperatures are too high and still need to be calibrated.

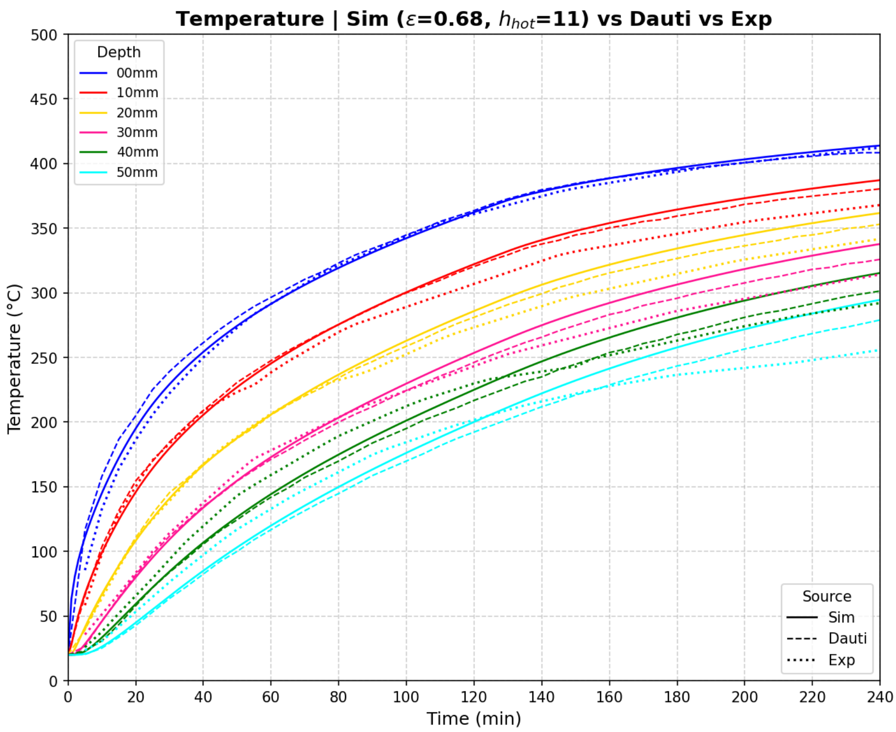
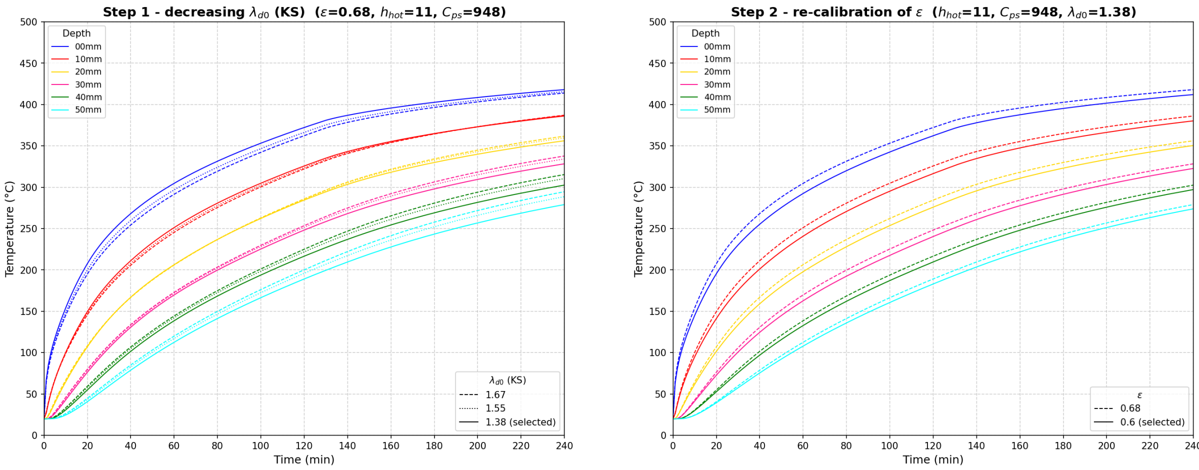


Figure 4.2: Temperature with the reference set ($\varepsilon = 0.68$, $h_c = 11$, $C_{ps} = 948$, $\lambda_{d0} = 1.67$), compared with the experiment and with Dauti's results.

Step 1 – reducing the conductivity λ_{d0} (KS). The conductivity acts mostly on the layers *closer to the heated face*. To bring down the over-predicted 10 and 20 mm curves, λ_{d0} is reduced (Figure 4.3a). This indeed lowers those curves, but it has a side effect: by slowing the inward transfer, heat accumulates near the surface, so the surface temperature becomes too high in the tail of the curve and, for the lowest values, the computation eventually crashes.

Step 2 – re-calibrating the emissivity ε. The surface overheating introduced by the lower λ_{d0} is corrected by re-calibrating the emissivity. Reducing ε lowers the radiative input at the surface (Figure 4.3b) and restores the 0 mm response, stabilising the run.

Step 3 – increasing the heat capacity C_{ps} (CS). The heat capacity governs the thermal inertia and is felt mostly at the *deepest points*. To bring down the still over-



(a) Effect of decreasing conductivity λ_{d0} (`KS`).

(b) Re-calibration of emissivity ε.

Figure 4.3: In-depth temperature calibration: Steps 1 and 2 to correct the outer layers and surface overheating.

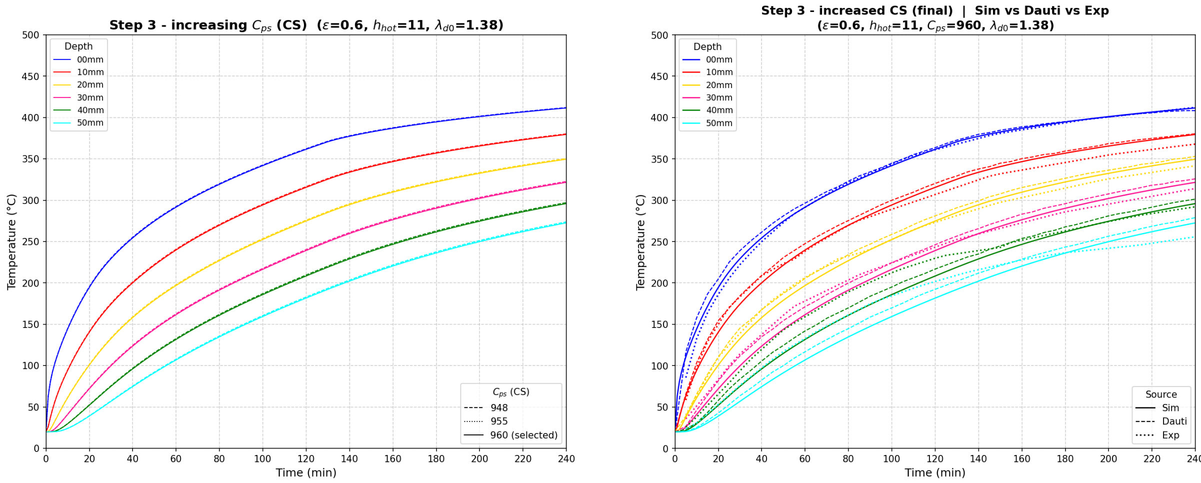
predicted 30, 40 and 50 mm curves, C_{ps} is increased (Figure 4.4a): a larger C_{ps} stores more energy per unit temperature rise, so the same heat flux produces a smaller, more delayed temperature increase in the core, lowering the internal temperatures.

Balancing C_{ps} and λ_{d0}. Increasing C_{ps}, however, cannot be pushed independently of λ_{d0}: the two parameters are coupled and must be balanced against each other. If C_{ps} is raised while λ_{d0} is kept low, the material both stores a large amount of energy and conducts it inward too slowly, so the incoming heat cannot diffuse away from the surface. This was observed directly for the combination $\varepsilon = 0.68$, $h^{\text{hot}} = 11$, $C_{ps} = 960$, $\lambda_{d0} = 1.38$: the surface heat could not pass inward, the surface temperature rose abnormally, and the run crashed. The combination only becomes stable once the surface input is reduced – by lowering the emissivity to $\varepsilon = 0.6$ (Step 2) – so that a high C_{ps} and a low λ_{d0} can coexist. The calibration therefore consists of finding a balanced (λ_{d0}, C_{ps}) pair, together with the matching surface input, rather than tuning either conduction parameter on its own.

At the end of this journey the temperature is reproduced consistently at all depths (0–50 mm). The final comparison against the Dauti and experimental curves is shown in Figure 4.4b, providing a reliable thermal loading for the gas-pressure calibration.

4.3 Gas-pressure calibration

With the temperature field fixed, the calibration of the gas-pressure evolution is addressed. The pressure build-up is governed primarily by how easily vapor can escape the heated region (permeability) and the amount of water available for vaporization.



(a) Effect of increasing heat capacity C_{ps} (CS). (b) Final calibrated temperature field.

Figure 4.4: In-depth temperature calibration: Step 3 and the final validation of the thermal model.

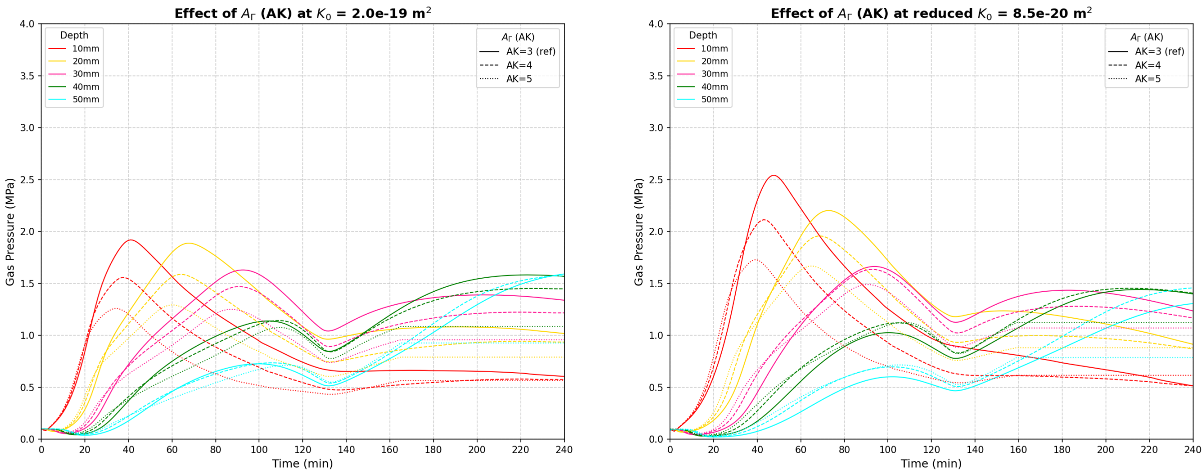
4.3.1 Primary parameters: K_0 and A_Γ

The intrinsic permeability K_0 (denoted KINT) and the permeability–hydration coefficient A_Γ (denoted AK) are the two most influential parameters dictating the magnitude and acceleration of the gas transport. In the newly implemented THC model, the evolution of permeability is highly sensitive to the degradation of the effective hydration degree $\tilde{\Gamma}$ **??**.

These two parameters must be strictly balanced. If the combination makes K too low, vapour cannot escape and the trapped vapor generates extremely severe pressure gradients; if it makes K too high, the vapor escapes too easily into the environment, resulting in flattened, underestimated pressure profiles. In both cases, the Newton-Raphson solver diverges (crashes), so K_0 and A_Γ have to be calibrated carefully together.

Starting point – $K_0 = 2 \times 10^{-19}$ m^2. The first sweep is performed at the initial reference permeability $K_0 = 2 \times 10^{-19}$ m^2, varying A_Γ over 3, 4 and 5 (with $A_\Gamma = 3$ as the initial reference); the result is shown in Figure 4.5a. Increasing A_Γ raises the peaks, but this K_0 value is not physically representative of HPC, and the curves still do not fit the experimental pressures.

Reducing the permeability – $K_0 = 8.5 \times 10^{-20}$ m^2. The permeability is therefore reduced to $K_0 = 8.5 \times 10^{-20}$ m^2, which is more consistent with a dense HPC, and the A_Γ sweep is repeated (Figure 4.5b) to assess the effect of lowering K_0. Trapping the gas more effectively raises the peaks, and the pair $K_0 = 8.5 \times 10^{-20}$ m^2, $A_\Gamma = 4$ is retained as the best compromise.



(a) $K_0 = 2 \times 10^{-19}$ m^2 (initial reference). (b) $K_0 = 8.5 \times 10^{-20}$ m^2 (reduced).

Figure 4.5: Effect of permeability-hydration coefficient A_Γ across different initial intrinsic permeabilities.

Comparison with the references. The retained pair is compared against the Dauti and experimental curves in Figure 4.6. The agreement is reasonable in the outer layer of 10 mm, but a structural difficulty appears with depth, as discussed below.

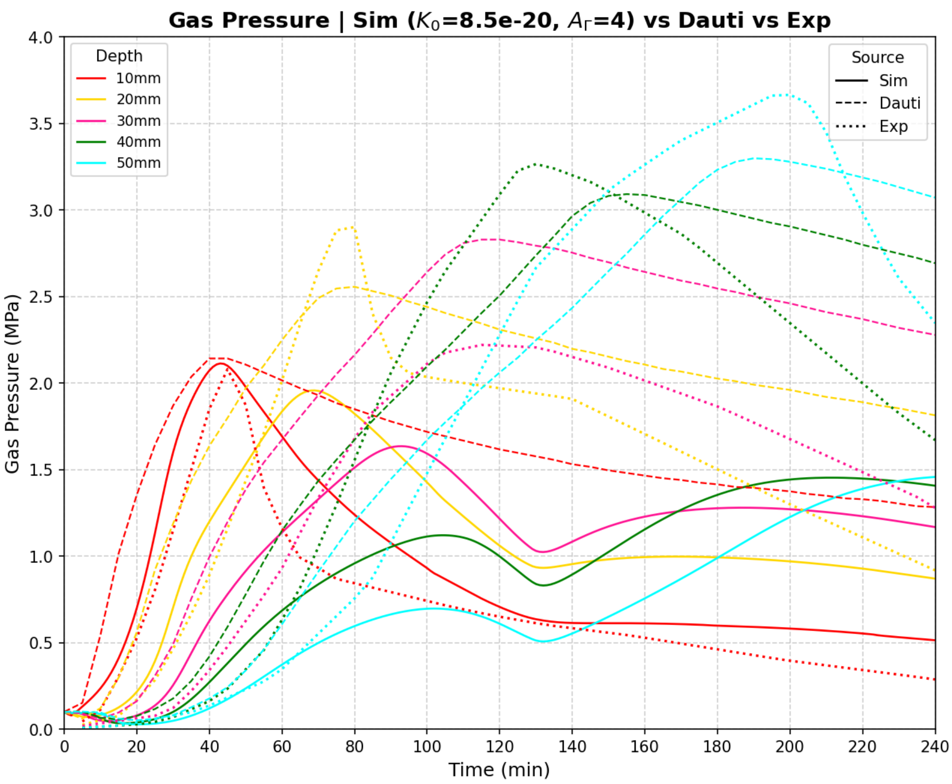


Figure 4.6: Gas pressure with the retained pair ($K_0 = 8.5 \times 10^{-20}$ m^2, $A_\Gamma = 4$): simulation (solid) vs. Dauti (dashed) vs. experiment (dotted)

The Main Issue: The sweep over (K_0, A_Γ) reveals the central difficulty of this stage: the gas pressure can be adjusted significantly only in the *outer layers* (10 and 20 mm). As one moves deeper into the specimen ($30, 40, 50$ mm), the simulated peak pressure *decreases*. This contradicts the experimental observations of Kalifa et al. [11], where deeper zones exhibited progressively higher peak pressures.

Physically, this discrepancy suggests that the high-gas-pressure region generated by the simulation is spatially smaller than it should be. The pressure build-up and the formation of the "moisture clog" remain concentrated too close to the exposed surface, failing to propagate massive condensation and pressurization into the deeper layers of the concrete core.

4.3.2 Secondary adjustments investigated

To address this depth-dependent pressure decay and force the moisture clog deeper into the material, several secondary parameters were investigated:

- **Initial Relative Humidity (RH_0):** While intuitively a higher initial saturation provides a larger water reservoir for vaporization, increasing RH_0 towards full saturation (e.g., > 0.99) induces severe numerical singularities in the capillary pressure gradients, systematically causing the code to crash.
- **Desorption Isotherm (a):** Adopting a stiffer desorption curve (higher a parameter, which is physically more appropriate for HPC to retain water at higher capillary pressures) successfully yielded a slight increase in the peak pressure at 20 mm, but failed to propagate this increment to the $30 - 50$ mm depths.
- **Aging / Curing time:** Modifying the hydration history significantly alters the initial state. Decreasing the aging time reduces the overall gas pressure and critically compromises the code's stability. On the other hand, increasing the aging time excessively subjects the HPC to severe self-desiccation, rendering the internal core unrealistically dry prior to heating.
- **Vapor mass-transfer coefficient (h_g):** In an attempt to artificially trap the gas, the convective exchange coefficient at the heated face was reduced to a minimum. While this approach successfully forced vapor inward and increased p_g in the deeper zones, it could only maintain numerical stability if compensated by an artificially large intrinsic permeability (K_0). However, imposing a large K_0 fundamentally contradicts the extremely dense, low-permeability nature of HPC containing silica fume.

The outcome of the last attempt is shown in Figure 4.7: with $h_g = 0.001$ and a large $K_0 = 2 \times 10^{-18}$ m^2, increasing A_Γ (1 to 4) does raise the deeper-zone pressures, but the overall pressure level is depressed because such an abnormally large K_0 lets the gas escape too easily.

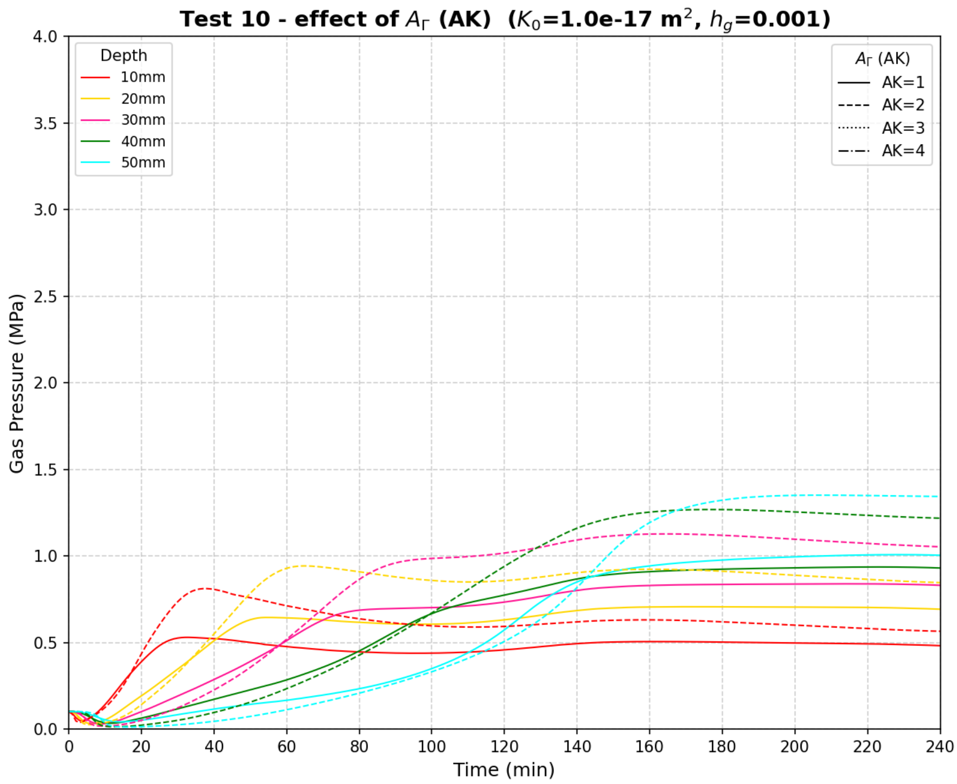


Figure 4.7: Gas-pressure histories at 10–50 mm for a minimal vapour transfer coefficient $h_g = 0.001$ and a large $K_0 = 2 \times 10^{-18}$ m^2, sweeping A_Γ from 1 to 4.

In summary, none of these secondary adjustments – alone or in combination with K_0 and A_Γ – could perfectly reproduce the deep pressure peaks without violating either the physical constraints of the material or the mathematical stability of the solver.

4.4 Validation and discussion

The reference model utilizes the calibrated parameter set optimized primarily for the thermal field and the near-surface gas pressure. Based on the calibration journey, the following critical discussions and validation outcomes are drawn.

Table 4.1: Final calibrated parameter set retained for the reference model.

Stage	Parameter	Retained value
Surface heat exchange	Convective coefficient h_c	$11\ \mathrm{W\,m^{-2}K^{-1}}$
	Emissivity ε	0.6
Bulk conduction	Reference dry conductivity λ_{d0}	$1.38\ \mathrm{W\,m^{-1}K^{-1}}$
	Solid specific heat C_{ps}	$960\ \mathrm{J\,kg^{-1}K^{-1}}$
Gas pressure	Intrinsic permeability K_0	$8.5 \times 10^{-20}\ \mathrm{m^2}$
	Permeability–hydration coeff. A_Γ	4

4.4.1 Temperature

The temperature validation is the in-depth comparison already obtained at the end of the calibration (Figure 4.4b): with the calibrated set of Table 4.1, the model reproduces the measured temperature histories at all depths within a few degrees, confirming that the thermal sub-problem and its boundary conditions are correctly captured. Importantly, the gas-pressure parameters K_0 and A_Γ calibrated in the last stage have a negligible effect on the temperature field, so the temperature agreement of Figure 4.4b remains valid for the final parameter set. The temperature field can therefore be used as a trustworthy loading for the hydric and mechanical sub-problems.

4.4.2 Gas pressure

The gas-pressure validation is the comparison obtained with the retained pair (Figure 4.6). The response **could not be validated**: even with the calibrated set of Table 4.1, and across the explored parameter space – K_0, A_Γ, RH_0, h_g, the desorption parameter a and the aging time – the model reproduces the pressure only in the outer layers (10 and 20 mm). In the deeper zones the simulated peak pressure keeps decreasing with depth, while the measurements show the opposite trend, so the high-pressure region remains too concentrated near the exposed surface.

Additionally, the conduction parameters C_{ps} and λ_{d0} retained for the temperature have a negligible effect on the gas pressure. With the fact K_0 and A_Γ ave a negligible effect on the temperature field, these justify the sequential calibration adopted in this chapter.

4.4.3 Dicussions

Two complementary explanations are proposed for this discrepancy.

- The first concerns the experiment itself. Measuring pore pressure requires inserting a gauge into the concrete, which inevitably creates a local discontinuity – a small cavity surrounded by an already porous and heterogeneous material. It is therefore not certain that these gauges record the true gas pressure of the intact porous network; part of what they measure may reflect the local conditions of the cavity and the surrounding damaged zone rather than the field the model computes. This raises a genuine question about what the deep-zone gauges are actually measuring, and hence about the reference against which the model is being judged.

- The second concerns the model. The formulation used here is the recent THC model, which is still being developed; its saturation, permeability and dehydration laws have not yet been fully tuned to the high-temperature behaviour of HPC. The fact that the deeper-zone pressures could be raised only through physically unreasonable parameter values (very high K_0, minimal h_g) suggests that the model is missing a mechanism that redistributes the pressure build-up further inward, and that further refinement of the THC formulation is required before a quantitative validation of the gas pressure can be expected.

Given the inherent uncertainties in both the non-linear material parameters and the ambiguous experimental gas-pressure data, relying solely on deterministic calibration is insufficient. This robustly justifies the necessity of shifting toward a probabilistic framework, utilizing uncertainty quantification to assess the spalling risk, which will be the core focus of Chapter 5.

Chapter 5

Stochastic assessment of the model

5.1 Sources of uncertainty and concrete heterogeneity

5.1.1 Motivation for probabilistic assessment

In civil engineering, a structure is considered safe when it remains fit for its intended use throughout its service life [2]. Deterministic models have traditionally been used to evaluate structural safety. However, the gap between a real physical system and a simplified numerical model, along with the unforeseeable nature of future loads and environmental conditions, inherently leads to uncertainty [1].

To manage these risks, modern engineering codes such as the Eurocodes adopt a probabilistic format [2]. This allows the safety of a structure to be explicitly quantified through its probability of failure P_f (Appendix B.6) and its reliability $R = 1 - P_f$. By introducing random variables into mechanical models and propagating them through structural analyses (Section 5.2), objective decisions can be made based on both the probability of an event and the severity of its consequences [1].

5.1.2 Classification of uncertainties

According to reliability theory, uncertainties can be classified into two categories [2, 1]:

- **Aleatoric uncertainties** arise from the natural randomness of the environment and the spatial or temporal variability of material properties. They cannot be reduced; they can only be quantified and accounted for in the design process [1].
- **Epistemic uncertainties** stem from incomplete knowledge or insufficient data. They include *model uncertainties* (discrepancy between mathematical models and

actual physical processes) and *statistical uncertainties* (limited or inaccurate data). Unlike aleatoric uncertainties, they can be reduced by improving models and gathering more data [1].

5.1.3 Concrete heterogeneity and material variability

A critical source of aleatoric uncertainty is the inherent heterogeneity of concrete [1, 7]. Its mechanical properties depend on the formation history, casting conditions, internal chemical reactions, and long-term degradation processes [7].

This variability manifests as *continuous* spatial fluctuations (e.g. Young's modulus, permeability, porosity) and *discrete* defects (e.g. micro-cracks, capillary networks) [1]. Consequently, material parameters must be modelled as random variables or random fields rather than deterministic values (Section 5.2).

5.2 Stochastic finite element framework

In structural engineering, material properties such as the Young's modulus, permeability, or thermal conductivity are never perfectly known. As noted by [7], even under strict quality control the compressive strength of high-performance concrete can range from 62 to 100 MPa. Two approaches exist to handle this uncertainty:

1. A *deterministic* analysis assigns a single fixed value to each parameter and produces a unique structural response; it cannot, by itself, quantify the likelihood that a design limit will be exceeded.
2. A *stochastic* analysis, on the other hand, treats uncertain parameters as random quantities, propagates this uncertainty through the model, and delivers statistical information—mean, variance, probability of failure—about the structural response.

To handle uncertainties in the finite element model, two levels of stochastic description are applied in this work. While various numerical tools are employed across the different analyses (e.g., LHS, Monte Carlo, PCE, KDE, Gaussian Quadrature), they all share a unified mathematical foundation rooted in the Probability Density Function (PDF) and the Cumulative Distribution Function (CDF). The full mathematical definitions linking these core probabilistic concepts to the utilized simulation methods are detailed in Appendix B.

Specifically, the roles of the PDF and CDF form a continuous stochastic workflow throughout the analyses:

- **Input:** The uncertainty of every material parameter (e.g., permeability, w/c) is initially defined by a prescribed PDF, indicating the most probable physical values.
- **Sampling (LHS & MC):** To perform probabilistic simulations, discrete values must be drawn from these distributions. While Monte Carlo (MC) draws randomly, Latin Hypercube Sampling (LHS) uses the CDF to divide the probability space into N equiprobable intervals, ensuring a space-filling design without clustering.
- **Propagation (PCE):** Since running thousands of non-linear FEM simulations is computationally prohibitive, Polynomial Chaos Expansion (PCE) acts as a mathematical filter, directly and rapidly transforming the input PDFs into the output PDF.
- **Output and Risk (KDE & Quadrature):** Once the structural responses are obtained, the probability of failure (P_f) must be evaluated. Kernel Density Estimation (KDE) smoothes the discrete sampled data into a continuous response PDF. Alternatively, the Gaussian Quadrature method computes the statistical moments to analytically reconstruct the response PDF. Finally, the probability of failure is evaluated directly by integrating the tail of this PDF over the failure domain (i.e., the shaded area beyond the critical threshold).

Within this general workflow, the spatial dimension of the uncertainty is treated at two levels:

Random Variable (RV). Each uncertain property is described by a single probability distribution applied uniformly over the whole specimen. This approach is used for the sensitivity (Section 5.5) and reliability (Section 5.6) analyses.

Random Field (RF). The property varies spatially across the domain, subject to a correlation structure. This level is used for the spatial heterogeneity study in Section 5.7.

[REVIEW] The following sections are currently under review and need to be rewritten

5.3 Uncertain inputs and quantity of interest

5.3.1 Two separate analyses

The uncertainty is propagated through *two independent studies* rather than a single combined one:

Analysis A — intrinsic permeability K_0. A *direct* uncertainty on a transport property, identified in Chapter 4 as the dominant control on the gas-pressure magnitude.

Analysis B — mix composition. An *upstream* uncertainty on the mix-design proportions, introduced at its physical source and propagated through Powers' model (Chapter 2, Appendix A), which converts the proportions into porosity, hydration degree, sorption and the resulting transport properties.

The two are kept apart for two reasons. First, they are of different nature — a material-property scatter on one side, an upstream design imprecision on the other — and mixing them would blur the interpretation. Second, K_0 is known from Chapter 4 to dominate the pressure response; pooling it with the four mix proportions in a single variance decomposition would let its variance swamp the Sobol indices (Appendix B.5.2) and *mask* the relative importance *among* the mix proportions. Separating the analyses lets each source be characterised on its own terms.

5.3.2 Choice of probability distributions

Three distribution families are used (Table 5.1), each chosen to reflect the *nature* of the corresponding uncertainty rather than by convenience; the formal definitions are given in Appendix B.1.

Lognormal — permeability K_0. The intrinsic permeability is strictly positive and varies by multiplicative factors: it is known only to within an order of magnitude and its scatter is naturally expressed as a ratio, not an additive offset. A lognormal law guarantees $K_0 > 0$ for every sample and reproduces this multiplicative, positively skewed variability; it is the standard model for the permeability of concrete [7, 16]. It is also consistent with the random-field generator of Section 5.7: the Cast3M `ALEA` operator produces a Gaussian field that is exponentiated into a lognormal one (Appendix B.2.4), so the random-variable and random-field descriptions of K_0 share the same marginal.

Normal — cement content c and aggregate content agg. The batched masses (in $\mathrm{kg\,m^{-3}}$) are controlled by the mixing plant around a target value. Their variability is the accumulation of many small, independent dosing and weighing errors, which by the central-limit theorem tends to a Gaussian centred on the nominal dosage. The scatter is small relative to the mean (CoV $\approx 4\%$ for c, $\approx 3\%$ for agg), so the symmetric Normal law describes the batch-to-batch dispersion faithfully and the probability of a physically meaningless (negative) sample is negligible.

Uniform — ratios w/c and s/c. For the water/cement and silica/cement ratios, what is known is not a measured central tendency but an *admissible design window*: an

HPC of this type is specified to lie within $w/c \in [0.30, 0.38]$ and $s/c \in [0.08, 0.12]$. When only the bounds are known and there is no basis to prefer any value inside them, the maximum-entropy (least-informative) choice is the uniform distribution [1]. It models the *epistemic* imprecision of the as-specified mix rather than a measurement scatter, and it keeps every sample strictly within the physically admissible window, avoiding tails that would push Powers' model or the solver outside its valid range.

The four mix proportions are taken as mutually independent design choices; their physical coupling (a given w/c and s/c jointly set porosity and permeability) is produced *inside* the model by Powers' stoichiometry, not imposed on the inputs. The OpenTURNS declaration is given in Table 5.1.

Table 5.1: Random inputs of the two analyses. calibrated K_0 and the mix proportions

Input	Symbol	Distribution	Parameters
Analysis A — transport property			
Intrinsic permeability	K_0	Lognormal	$\mu = 8.5 \times 10^{-20}$ m^2, CoV = 0.30
Analysis B — mix composition			
Cement content	c	Normal	$\mu = 377,\ \sigma = 15$ kg m^{-3}
Aggregate content	agg	Normal	$\mu = 1920,\ \sigma = 50$ kg m^{-3}
Water/cement ratio	w/c	Uniform	$[0.30,\ 0.38]$
Silica/cement ratio	s/c	Uniform	$[0.08,\ 0.12]$

5.3.3 Quantity of interest and limit state

Both analyses share the same scalar quantity of interest (QoI): the peak gas pressure at the 10 mm depth:

$$p_g^{10} = \max_t p_g(x = 10 \text{ mm}, t). \tag{5.1}$$

Following the general reliability framework defined in Appendix B.6.1, the specific limit-state function for this study is defined as:

$$G(\mathbf{X}) = p_{\text{crit}} - p_g^{10}(\mathbf{X}), \tag{5.2}$$

where $\mathbf{X}$ is the input vector of the analysis considered and p_{crit} is the critical pressure tied to the temperature-degraded tensile strength. Failure corresponds to $G(\mathbf{X}) \leq 0$.

5.4 Uncertainty propagation with OpenTURNS

Uncertainty quantification is carried out with the OpenTURNS library [**baudin2017**], reusing the Python automation layer built for the calibration sweeps of Section 4.1.2. There, `itertools.product` is also reused for a probabilistic design.

Besides, extra steps is required for the automated stochastic analyses:

1. Define the distribution(Table 5.1), using `ComposedDistribution`.
2. Generate an LHS design of N samples (**??**).
3. Python script for post-process and ploting: correlation, Sobol indices, response PDF.

To reduce computational cost compared to standard Monte Carlo, Latin Hypercube Sampling (LHS) is employed via OpenTURNS. The theoretical details of LHS are presented in Appendix B.3.2.

5.5 Sensitivity analysis

5.5.1 Analysis A — permeability K_0

With A_Γ held constant to isolate the magnitude-driving uncertainty, Analysis A evaluates how p_g^{10} responds to K_0. Since there is only one random variable, a Sobol decomposition is not meaningful ($S_1 = S_T = 1$). The results highlight a strongly negative correlation: lower permeability traps more vapour, which results in a higher pressure peak (Figures 5.1 and 5.2).

For the interpretation of the transient response plots and sensitivity bar charts in this chapter, it is important to clarify the terminology and statistical metrics used:

- **Sim vs. Baseline:** In the legends, **Sim** (or Sim Mean) represents the mean of all stochastic simulations evaluated with OpenTURNS (reflecting the full propagated probability distribution), whereas the **Baseline** corresponds to the single deterministic simulation utilizing the optimal parameter set already calibrated in Chapter 4.
- **95% Confidence Interval (CI) Bands:** The shaded areas around the transient curves and the error bars on the Sobol indices represent the 95% confidence intervals. For the pressure histories, this band illustrates the physical dispersion, indicating a 95% probability that the pressure response falls within this zone under the given

input uncertainties. For the Sobol indices, it represents the statistical bootstrap estimation error due to a finite sample size.

The comprehensive theoretical background regarding the stochastic framework, probability density functions, and the required number of evaluations for reaching a 95% confidence level is detailed in Appendix B.3.

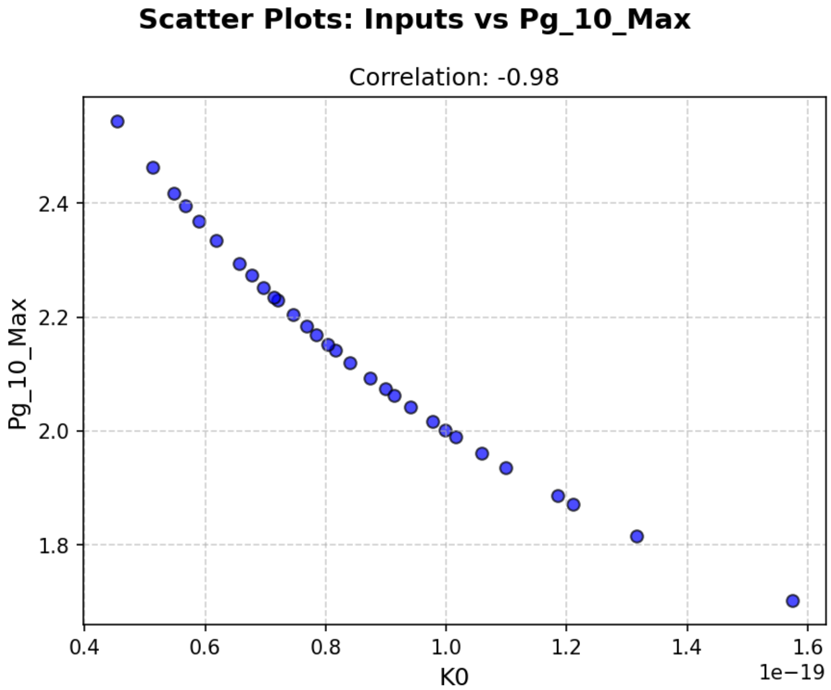


Figure 5.1: Analysis A— correlation: Pearson and Spearman rank correlation of K_0 with the peak pressure p_g^{10}.

5.5.2 Analysis B — mix composition

In Analysis B, the four mix proportions are treated as random variables. To overcome the computational bottleneck, a surrogate model is constructed within OpenTURNS using Polynomial Chaos Expansion (PCE). The variance-based Sobol indices (see theoretical definitions in Appendix B.5.2) are then extracted analytically from the metamodel, identifying the dominant parameter group and their cross-interactions (Figures 5.3 to 5.5).

Conclusion. In Analysis A Figure 5.1 the permeability shows a clear, almost deterministic dependence: K_0 is strongly and negatively correlated with p_g^{10} (Pearson/Spearman ≈ -0.98), the points falling on a tight monotone curve, which confirms permeability as the controlling input. In Analysis B Figure 5.3 the picture is far more scattered: apart from w/c, which is strongly correlated with p_g^{10} (≈ 0.93), only the cement content shows an appreciable correlation (≈ 0.34), while the aggregate content and s/c are essentially uncorrelated (-0.11 and 0.24). The variance-based ranking confirms and sharpens this:

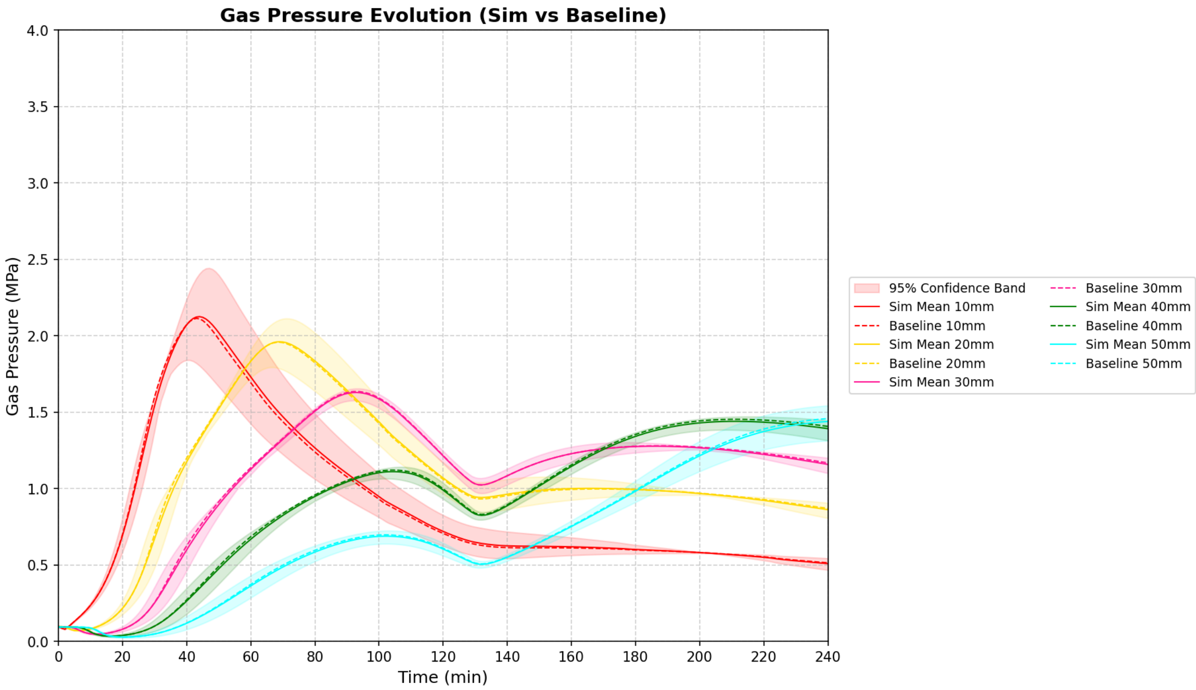


Figure 5.2: Analysis A — bootstrap 95% confidence-interval bands of the first- and total-order Sobol indices.

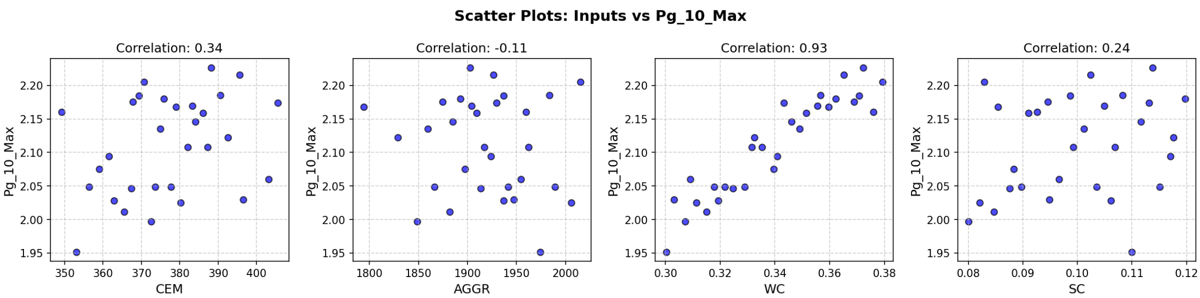


Figure 5.3: Analysis B — correlation: Pearson and Spearman rank correlation of each mix proportion $(c, agg, w/c, s/c)$ with the peak pressure p_g^{10}.

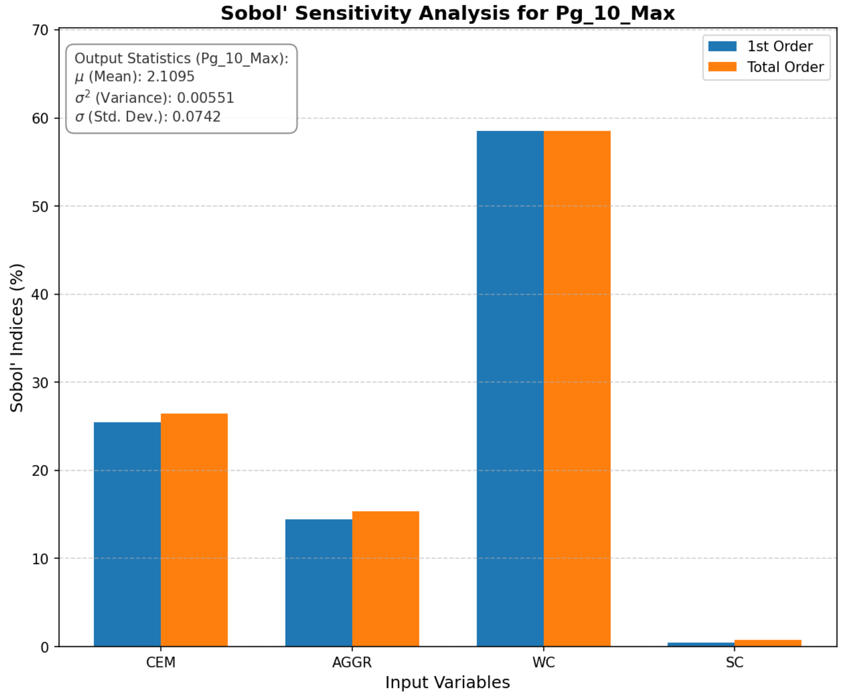


Figure 5.4: Analysis B — first- and total-order Sobol indices (S_i, S_{T_i}) of the mix proportions on p_g^{10}.

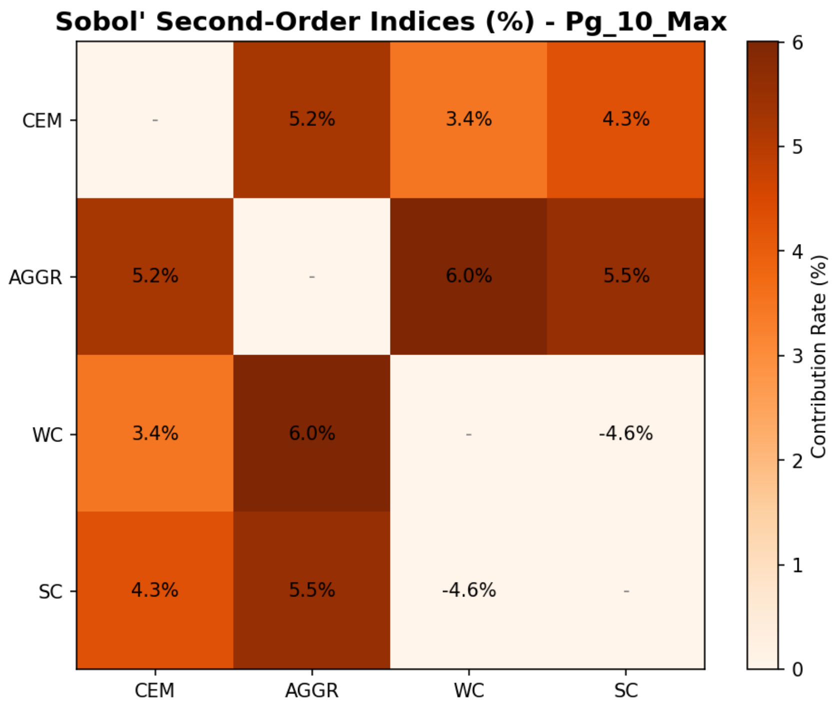


Figure 5.5: Analysis B — second-order Sobol indices S_{ij} of the mix-proportion pairs on p_g^{10}.

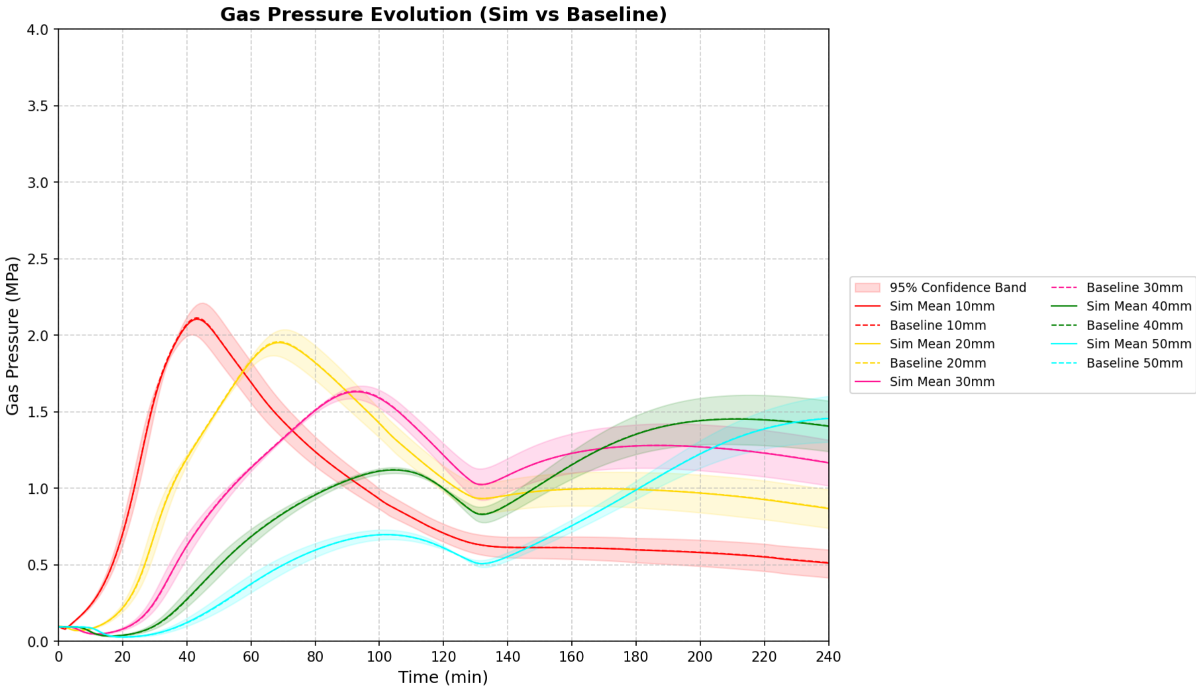


Figure 5.6: Analysis B — bootstrap 95% confidence-interval bands of the first- and total-order Sobol indices.

w/c carries by far the largest Sobol index ($S_i \approx S_{T_i} \approx 0.58$), followed by the cement content (≈ 0.25) and the aggregate content (≈ 0.15), while s/c is negligible (< 0.01). The first- and total-order indices nearly coincide for every input, so the interactions are weak and the response is essentially additive, and the second-order indices (Figure 5.5) are correspondingly small. Notably, the aggregate content is almost uncorrelated yet still carries a non-negligible total index, a contribution that the monotone correlation misses but the variance decomposition captures.

The dominance of w/c shows that the initial free-water content of the mix governs the peak pressure far more than any other proportion: a higher w/c means both a more open pore structure and more capillary water available to vaporise, and both feed the gas pressure. This is fully consistent with Section 4.3.2, where a higher initial relative humidity was found to release more vapour and raise p_g^{10}.

Finally, the mix-induced scatter is small in absolute terms (mean $p_g^{10} = 2.11$ MPa, standard deviation 0.074 MPa), which is why Analysis B yields a negligible failure probability in Section 5.6; the proportions with a negligible total index, s/c in particular, can be fixed at their mean there.

5.6 Reliability analysis

The propagated samples yield a discrete set of structural responses. Using the continuous Probability Density Function (PDF) reconstruction methods available in OpenTURNS, we can quantify the spalling risk. Based on the definitions in Appendix B.6, the probability of failure P_f is evaluated.

Depending on the stochastic method employed, two distinct numerical strategies are utilized to reconstruct the distributions and evaluate the failure probability:

Kernel Density Estimation (Analyses A, C1, and C2): For analyses where the input uncertainty is limited to a single random variable (e.g., global K_0) or governed by a spatial random field (`ALEA`), building a multidimensional metamodel is either mathematically unnecessary or algorithmically incompatible. Instead, the continuous PDF is reconstructed directly from the raw, macroscopic Cast3M outputs using non-parametric Kernel Density Estimation (KDE). The probability of failure is then directly estimated by integrating the tail of this smoothed empirical distribution.

Monte Carlo on PCE Metamodel (Analysis B): In the multivariable mix-composition study, evaluating the failure probability directly from the limited 30-sample LHS design would yield highly inaccurate tail estimations. Instead, a massive Monte Carlo simulation (10^5 realizations) is executed exclusively on the Polynomial Chaos Expansion (PCE) surrogate model—which was previously constructed for the sensitivity analysis (Section 5.5.2). This allows for a highly accurate and computationally inexpensive evaluation of P_f without requiring additional Cast3M runs.

These reconstructed PDFs (Figures 5.7 and 5.8) visually demonstrate how the input uncertainties propagate towards the critical threshold.

Conclusion. With the critical threshold set at $p_{\text{crit}} = 2.5$ MPa:

- **Analysis A (Figure 5.7):** The peak pressure spreads widely, and its upper tail crosses the threshold, giving a failure probability $P_f = 6.59\%$.
- **Analysis B (Figure 5.8):** The distribution is narrower (confined between 1.8 and 2.4 MPa), meaning no sampled realisation reaches p_{crit} ($P_f \approx 0$).

This confirms that the intrinsic permeability K_0 is by far the dominant source of spalling risk.

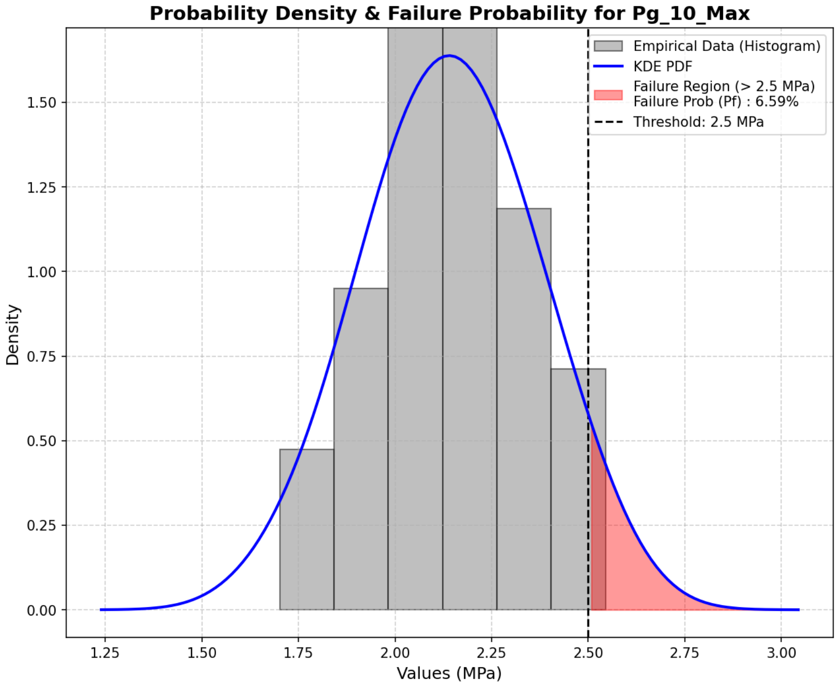


Figure 5.7: Analysis A: PDF of the peak pressure p_g^{10} under the permeability uncertainty (K_0), with the critical threshold p_{crit} (shaded tail $= P_f$) and the measured Kalifa 10 mm peak.

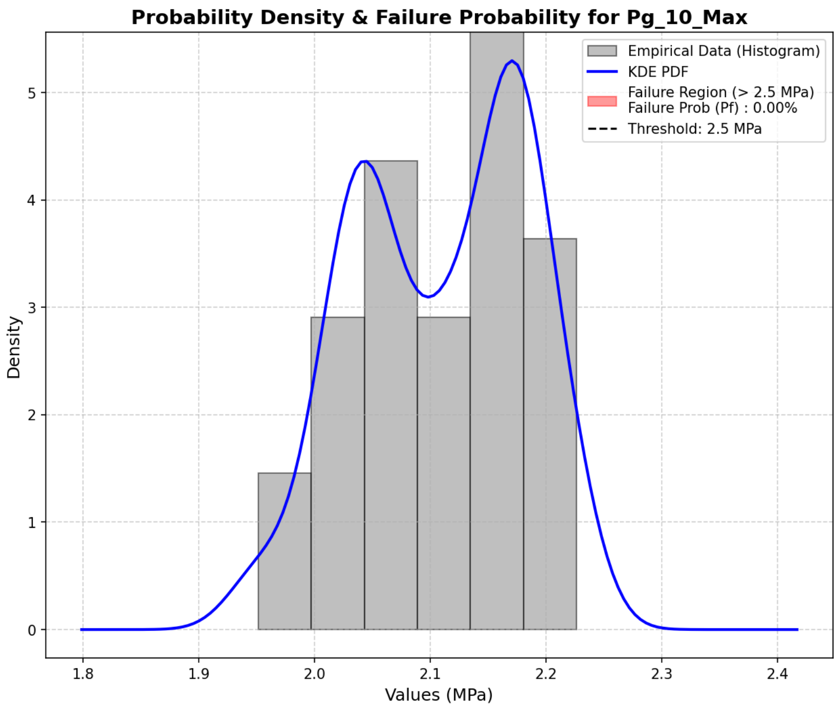


Figure 5.8: Analysis B: PDF of the peak pressure p_g^{10} under the mix-composition uncertainty $(c, agg, w/c, s/c)$, with the critical threshold p_{crit} (shaded tail $= P_f$) and the measured Kalifa 10 mm peak.

5.7 Effect of spatial heterogeneity on structural safety

The analyses above treat the permeability as a single random variable, constant over the specimen in a given realisation. Real concrete varies from point to point, and local weak zones — not average properties — govern failure. The intrinsic permeability K_{int}, the most spatially variable and most influential property (Section 5.5.1), is therefore additionally described as a *random field* and compared with the homogeneous description. The same lognormal marginal as in Analysis A is used (mean $= K_0^{\mathrm{cal}} = 8.5 \times 10^{-20}$ m^2, CoV $= 0.30$, calibrated in Chapter 4).

5.7.1 A simplified isothermal poro-mechanical model

Propagating a spatially heterogeneous random field through the fully coupled, highly non-linear Thermo-Hygro-Chemical (THC) model presents severe computational challenges. To obtain a controlled and tractable comparison of the stochastic methods, a *simplified isothermal poro-mechanical model* applied to a $10 \times 10 \times 10$ cm concrete cube is adopted instead. By exploiting symmetry, only 1/8 of the domain is modelled.

By holding the temperature constant at room conditions ($T = 20$°C), the complex thermo-chemical constitutive evolution is frozen. In this test case, the specimen is subjected to convective drying, where the internal relative humidity drops from an initial state of 90% to an environmental humidity of 50% at the exposed surfaces.

The intrinsic permeability K_0 is designated as the spatially variable input. The primary output of interest is the resulting *longitudinal shrinkage strain* induced strictly by capillary forces ($p_s = -S_l(P_c)P_c$). To avoid cluttering the main text, the detailed constitutive equations, mesh properties, and deterministic material parameters (e.g., Young's modulus, van Genuchten coefficients) of this simplified model are thoroughly documented in Appendix B.7.

5.7.2 Stochastic propagation methods

The uncertainty of the intrinsic permeability (K_0) is propagated using three distinct stochastic approaches, all detailed in Appendix B:

Analysis C1 (OpenTURNS + LHS): Homogeneous RV baseline (30 samples).

Analysis C2 (Cast3M `ALEA`): Heterogeneous Random Field using the Turning Bands Method with a spatial correlation length $L_c = 0.02$ m (30 samples).

Analysis C3 (Gaussian Quadrature): Homogeneous deterministic integration using `QUADRATU` (7 points).

The theoretical background for the TBM and Gaussian Quadrature frameworks can be found in Appendix B.2.3 and Appendix B.4, respectively.

However, for Analysis C3, because Cast3M lacks native support for exporting these continuous PDF curves for high-quality visualization, statical moments calculation and the Pearson type III reconstruction based on the quadrature moments was re-implemented in Python. This ensures consistent plotting and comparison against the LHS and random field methods. Ultimately, this approach serves as a computationally inexpensive cross-check to validate the statistical moments obtained from the LHS baseline (C1).

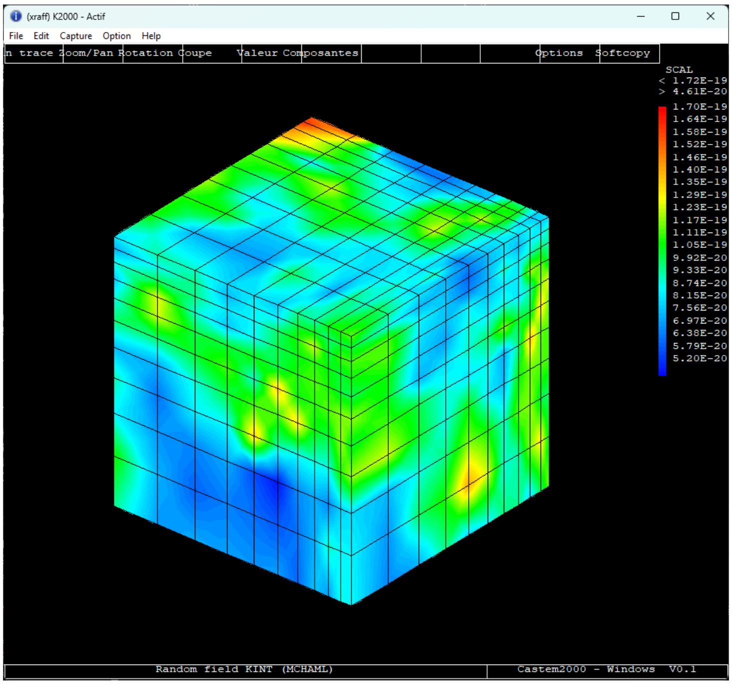


Figure 5.9: A single realization of the spatially heterogeneous intrinsic permeability field $K_0(\mathbf{x})$ generated on the 3D cube mesh using the Turning Bands Method ($L_c = 0.02$ m).

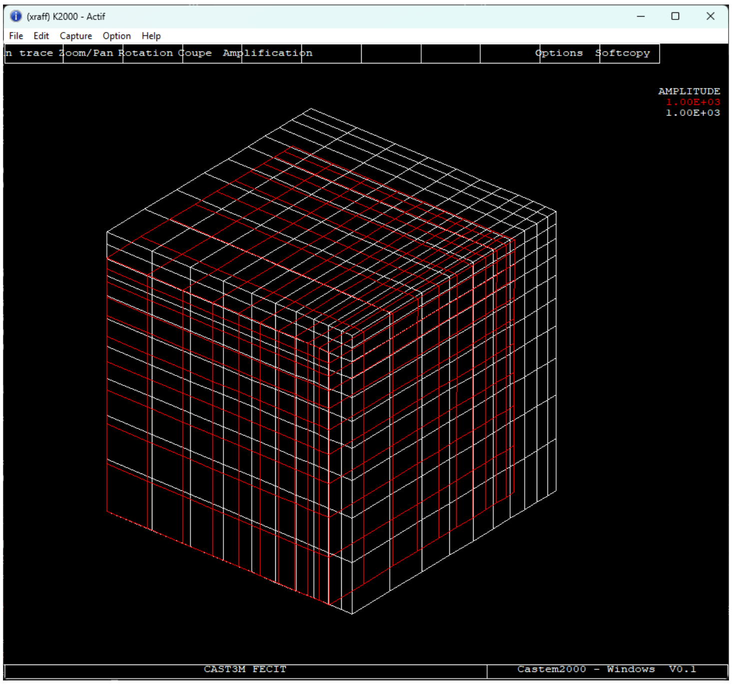


Figure 5.10: Resulting final shrinkage strain field of the concrete cube induced by capillary forces, highlighting the localized deformations driven by the heterogeneous permeability field.

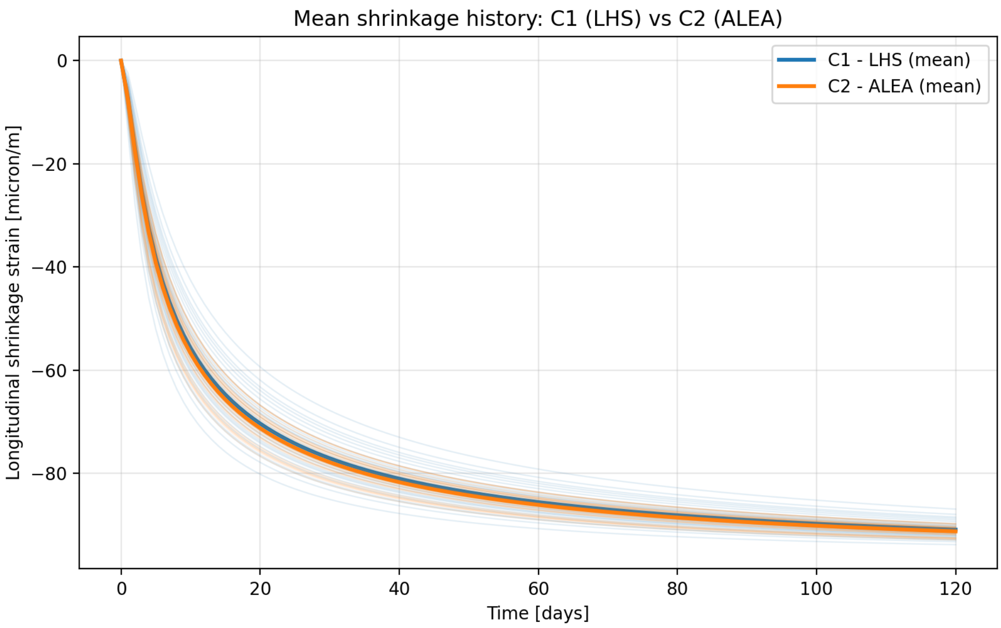


Figure 5.11: Mean longitudinal shrinkage strain history at the corner point for the two stochastic analyses, LHS (C1) and the spatial random field (C2). Faint lines show the individual realisations of each analysis.

The Figure 5.11 compares the mean shrinkage history of the two analyses before each is examined in detail. The ALEA mean (C2) lies slightly below the LHS mean (C1) throughout the drying period, while its realisations are visibly less scattered. These two features pull in opposite directions: a lower mean points to a more severe average response, but a tighter spread limits how often large strains are reached.

The mean curve alone therefore cannot identify the more critical analysis. This requires comparing the full distributions of the final strain, given by the probability density functions in Figure 5.13 and Figure 5.14. As illustrated in these figures, the significant gap between the homogeneous and heterogeneous results quantitatively highlights how a simple random-variable assumption dangerously underestimates the structural risk.

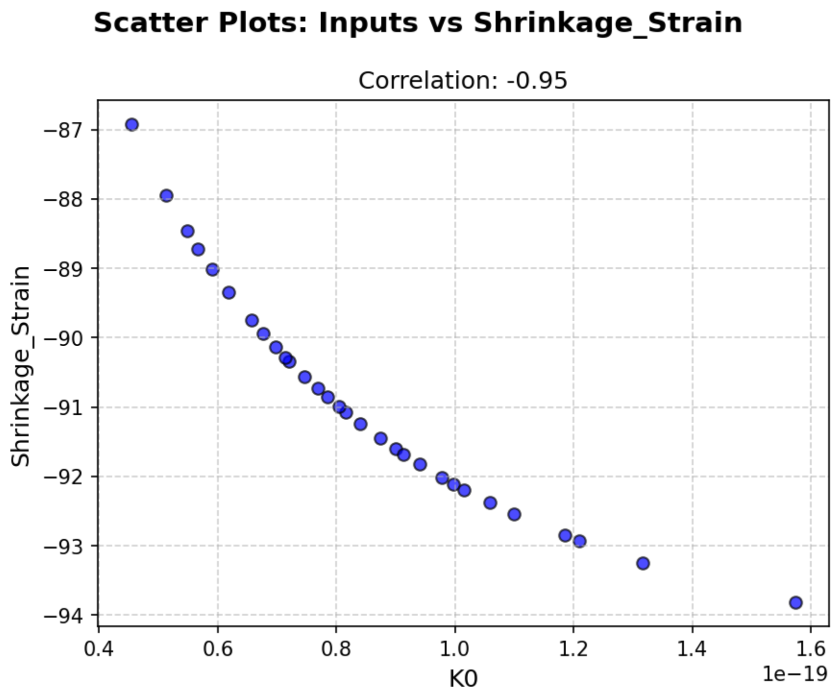


Figure 5.12: Analysis C1 (OpenTURNS + LHS): Pearson and Spearman rank correlation between the intrinsic permeability (K_0) and the resulting longitudinal shrinkage strain.

Conclusion. Two cross-checks and one primary physical observation are drawn from this comparative study. First, while both LHS (Analysis C1) and Gaussian quadrature (Analysis C3) treat the intrinsic permeability as a homogeneous random variable, their resulting probability distributions exhibit noticeable differences. Although the 7-point quadrature scheme provides a computationally inexpensive estimation, it yields a *more critical* response profile for the capillary pressure P_c and the longitudinal shrinkage strain compared to the 30-sample LHS baseline.

Second, and most importantly, the spatially heterogeneous random-field approach (Analysis C2) produces the *most critical* structural response. Physically, local low-permeability zones restrict moisture transport and amplify local capillary forces, which severely concentrates the deformation. This mechanism generates significantly higher peak shrinkage strains and a wider probability tail than both homogeneous cases (C1 and C3), because the highly stressed local weak zones cannot be entirely compensated by the surrounding material.

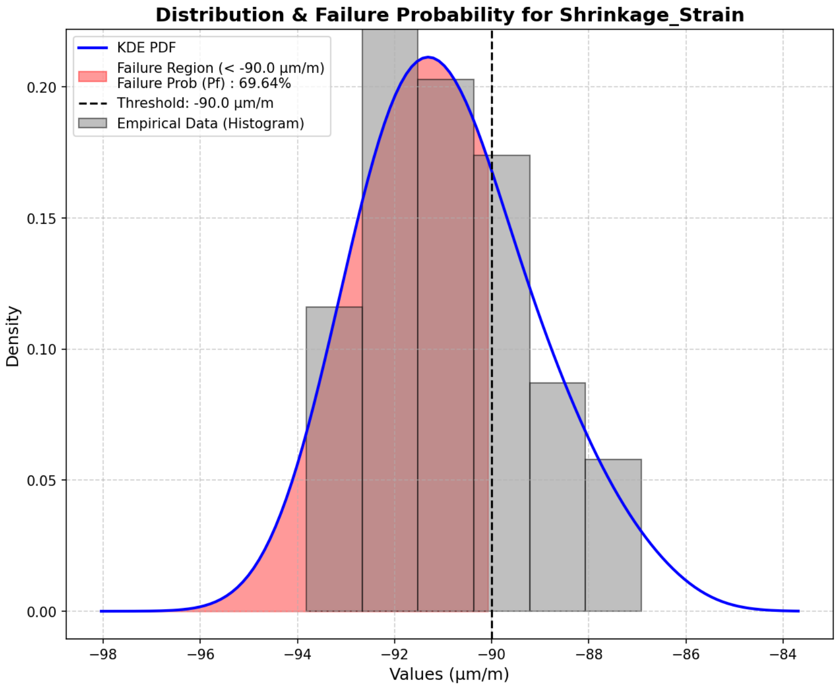


Figure 5.13: Analysis C1 (OpenTURNS + LHS): Probability Density Function (PDF) of the longitudinal shrinkage strain induced by the uncertainty of the intrinsic permeability (K_0).

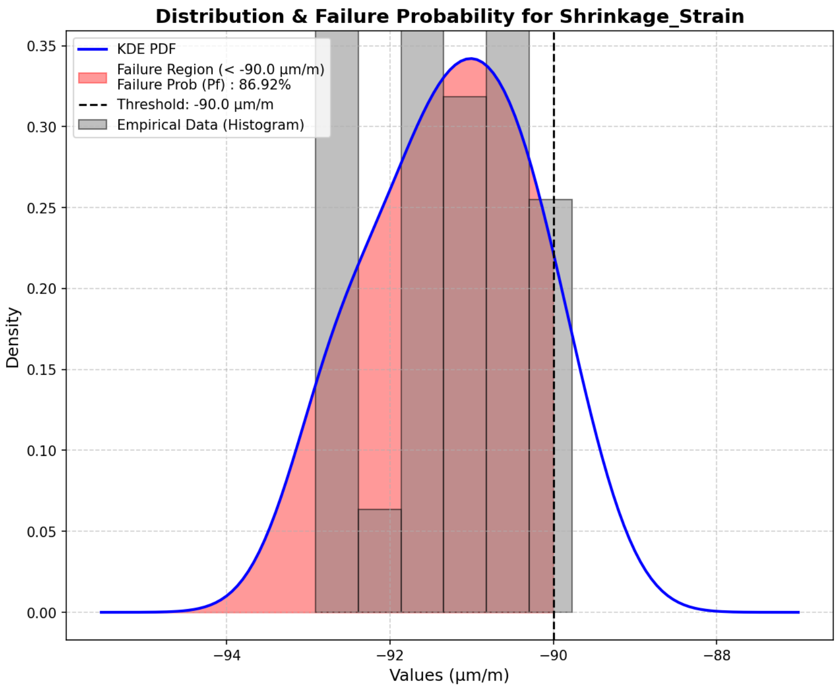


Figure 5.14: Analysis C2 (Cast3M `ALEA`): Probability Density Function (PDF) of the longitudinal shrinkage strain considering the spatial heterogeneity of the intrinsic permeability field.

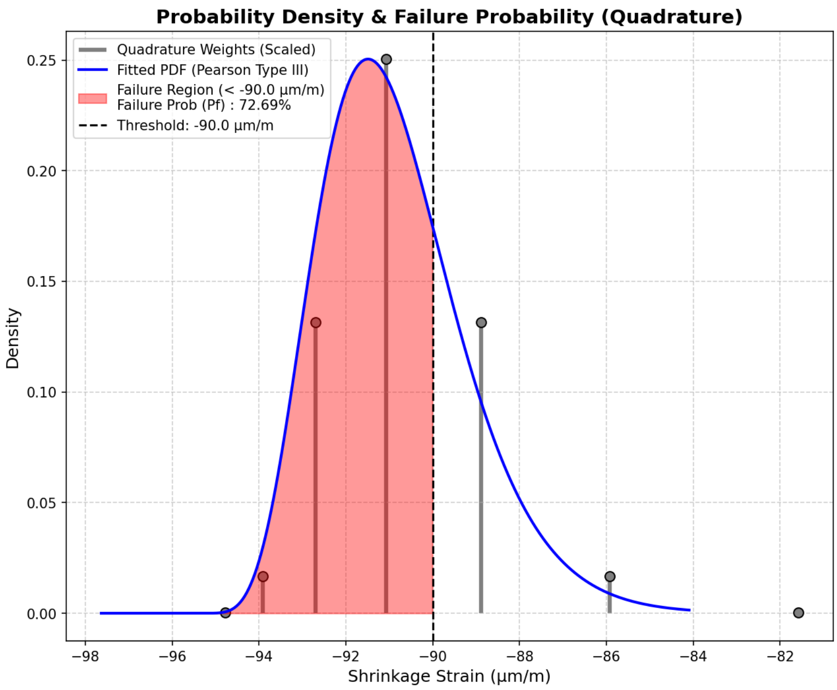


Figure 5.15: Analysis C3 (Gaussian Quadrature): Probability Density Function (PDF) of the longitudinal shrinkage strain evaluated through the deterministic few-point integration scheme.

The 30-realisation ensemble used for the Monte Carlo/LHS and random field analyses is a pragmatic estimate given the computational cost of each non-linear run; its scatter indicates the residual statistical uncertainty. Ultimately, the significant gap between the homogeneous and heterogeneous results quantitatively highlights how a simple random-variable assumption dangerously underestimates the structural risk. The full Turning Bands / `ALEA` construction, the quadrature operators, and the Monte Carlo convergence details are collected in Appendix B.

Chapter 6

Conclusions and Perspectives

[REVIEW] This chapter is under review

6.1 Main findings

The research conducted in this study has led to several key findings regarding the numerical modelling and stochastic assessment of high-performance concrete under severe accident conditions:

1. **Implementation of the THC model:** The improved Thermo-Hydro-Chemical (THC) model, which explicitly incorporates the effective hydration degree $\tilde{\Gamma}$ as an internal state variable, was successfully implemented and executed in CAST3M. This approach elegantly replaced empirical temperature-dependent multipliers with a strictly chemo-physical description of porosity and permeability evolution.

2. **Temperature calibration:** The automated calibration workflow successfully reproduced the temperature field. The simulated thermal profiles match the experimental data from the Kalifa test with high accuracy across all depths.

3. **Gas pressure calibration and its limits:** While the gas pressure could be calibrated reasonably well in the outer layers (10 and 20 mm), the model failed to capture the pressure distribution in the deeper zones regardless of the parameter adjustments (e.g., K_0, A_Γ, h_g). This discrepancy raises fundamental questions about both the completeness of the purely THC mathematical formulation and the intrusive nature of the experimental gauges.

4. **Sensitivity and dominant parameters:** A comprehensive stochastic assessment using OPENTURNS allowed for the variance-based sensitivity analysis of the model.

The extraction of first- and second-order Sobol indices identified the intrinsic permeability K_0 as the absolute dominant parameter governing the magnitude of the pore pressure, while mix-composition uncertainties presented negligible interaction effects on the pressure peaks.

5. **Reliability and spalling risk:** By propagating the material uncertainties, the reliability analysis successfully reconstructed the Probability Density Functions (PDFs) of the structural response. It quantified the probability of failure (P_f) and the risk of spalling based on a critical pressure threshold at the 10 mm layer, confirming that deterministic approaches alone are insufficient for structural safety assessment.

6. **Impact of spatial heterogeneity:** Using a simplified isothermal poro-mechanical model, the investigation into the spatial variability of permeability using random fields (`ALEA`) revealed that assuming a homogeneous material property dangerously underestimates the structural risk. Localized weak zones trigger more severe capillary shrinkage and pressure localizations compared to the averaged homogeneous baseline.

6.2 Model limitations

While the presented framework provides a deep insight into concrete behavior at high temperatures, several limitations remain:

- **Restriction to the THC scope:** Although the full THM framework was theoretically formulated, the numerical simulations for the high-temperature fire scenario were restricted to the THC scope. The absence of the mechanical module means that thermally and hydrally induced damage (*micro-cracking*) is not evaluated. Consequently, the damage-induced increase in permeability (M $\rightarrow$ H coupling) is missing, which likely explains the model's inability to relieve the accumulated gas pressure in the deeper layers.

- **Computational cost of Random Fields:** Propagating a highly refined spatial random field through the fully coupled, highly non-linear THC model proved to be computationally prohibitive. Therefore, the spatial heterogeneity study was constrained to a simplified poro-mechanical model rather than the full high-temperature accident scenario.

- **Ambiguity of experimental reference:** The intrusive nature of inserting pressure gauges into the concrete matrix creates artificial local cavities. It remains highly uncertain whether the deep-zone measurements truly reflect the intact pore pressure or the localized artifacts of the damaged zone surrounding the sensor.

6.3 Perspectives for future research

Building upon the limitations and findings of this work, several avenues for future research are identified:

- **Fully coupled THM formulation with micro-cracking feedback (M $\rightarrow$ H):** To resolve the gas pressure discrepancies at deep zones, the mechanical module must be fully integrated. By explicitly modelling *micro-cracking* (e.g., via the Mazars damage model) and its direct feedback on the intrinsic permeability, the simulation could capture the drastic transport acceleration that relieves internal pressures, providing a complete picture of the spalling mechanism.
- **Advanced surrogate modelling:** To make the stochastic propagation of random fields computationally tractable for the full highly non-linear THC solver, advanced metamodels (e.g., Polynomial Chaos Expansions or Artificial Neural Networks) should be developed. This would allow replacing the expensive Monte Carlo simulations with rapid analytical evaluations.
- **Non-intrusive experimental measurements:** To resolve the ambiguity of the gas-pressure measurements, future experimental campaigns should rely on non-intrusive techniques, such as in-situ neutron tomography, to observe the actual moisture migration and pressure redistribution without disturbing the concrete matrix.
- **Full service-life simulation:** Following the complete vision of modern frameworks, the THC formulation should be expanded to simulate the entire lifespan of the structure continuously—from early-age curing, through long-term aging and self-desiccation, up to the accidental fire scenario.
- **Broader verification:** Finally, the robustness of the model should be verified against a wider range of concrete mixtures (e.g., ordinary concrete, fiber-reinforced concrete) and under more severe heating scenarios, such as the ISO 834 or hydrocarbon fire curves.

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Appendix A

Thermo-Hydro-Mechanical behavior of Heated Concrete

[REVIEW] This chapter is under review

A.1 Thermal behavior

A.1.1 Conduction, convection and radiation

Heat transfer is the transport of thermal energy driven by a temperature difference. The following definitions of heat transfer modes are summarized from standard references [3].

Conduction

Conduction is the transfer of energy from the more energetic to the less energetic particles of a substance

- **Physical Mechanism:**
- **Rate Equation (Fourier's Law):** For a one-dimensional plane wall having a temperature distribution $T(x)$, the rate equation is expressed as Fourier's law:

$$q''_{cond} = -\lambda \frac{dT}{dx} \tag{A.1}$$

  Where q''_{cond} is the heat flux (W/m^2), λ is the thermal conductivity $(W/m \cdot K)$, and the minus sign indicates that heat is transferred in the direction of decreasing temperature.

Convection

The convection heat transfer mode is

- **Physical Mechanism:** Convection heat transfer occurs between a fluid in motion and a bounding surface when the two are at different temperatures
- **Rate Equation (Newton's Law of Cooling):** Regardless of the nature of the convection process, the appropriate rate equation is:

$$q''_{conv} = h(T_s - T_\infty) \tag{A.2}$$

  Where q'' is the convective heat flux (W/m^2), h is the convection heat transfer coefficient $(W/m^2 \cdot K)$, T_s is the surface temperature, and T_∞ is the fluid temperature.

Radiation

Thermal radiation is energy emitted by matter that is at a nonzero temperature, and unlike conduction and convection, it does not require the presence of a material medium.

- **Physical Mechanism:** The energy of the radiation field is transported by electromagnetic waves (or photons)
- **Emissive Power (Stefan-Boltzmann law):** The upper limit to the emissive power (rate of energy released per unit area) for an ideal radiator or blackbody is given:

$$E_b = \sigma T_s^4 \tag{A.3}$$

  The heat flux emitted by a real surface is given by $E = \varepsilon\sigma T_s^4$, where σ is the Stefan-Boltzmann constant and ε is the emissivity.
- **Net Radiation Exchange:** For a gray surface at T_s completely surrounded by a much larger isothermal surface at T_{sur}, the net rate of radiation heat transfer per unit area is:

$$q''_{rad} = \varepsilon\sigma(T_s^4 - T_{sur}^4) \tag{A.4}$$

A.1.2 Heat transfer

Heat Equation

Based on the principle of energy conservation and Fourier's law, the general heat diffusion equation (or heat equation) for a medium is detailed in [8] and can be expressed as:

$$\rho c_p \frac{\partial T}{\partial t} + \operatorname{div}(-\lambda \overrightarrow{\operatorname{grad}}(T)) - q = 0 \tag{A.5}$$

Where:

- ρ is the volumetric mass density (kg/m^3).
- c_p is the specific heat capacity ($J/kg \cdot K$).
- λ is the thermal conductivity ($W/m \cdot K$).
- T is the temperature field and t is time.
- q is the volume heat flux source (W/m^3).

To solve this equation, appropriate boundary conditions must be specified:

- **Prescribed temperatures :** The temperature is specified as $T = T_{imp}$ on the boundary surface ∂V^T.
- **Prescribed surface heat flux :** The total heat flux due to conduction, convection, and radiation on the boundary surface ∂V^φ is given by:

$$\vec{n} \cdot (\lambda \overrightarrow{\operatorname{grad}}(T)) = \varphi_{imp} + h(T_f - T) + \varepsilon\sigma(T_\infty^4 - T^4) \tag{A.6}$$

  where φ_{imp} is the prescribed heat flux, $h(T_f - T)$ represents the convection term, and $\varepsilon\sigma(T_\infty^4 - T^4)$ is the radiation term.

Thermal properties

The liquid water density evolves with temperature according to a polynomial fit [**furbish1997**]:

$$\rho_l(T) = \sum_{i=0}^{5} a_i T^i \tag{A.7}$$

with coefficients a_i given in Table **??**.

The solid thermal capacity theoretically evolves linearly with temperature [**harmathy1973**]:

$$C_{ps}(T) = C_{ps0}[1 + A_c(T - T_0)] \tag{A.8}$$

with $C_{ps0} = 940\ \mathrm{J\,kg^{-1}\,K^{-1}}$ and A_c is a material constant.

Note on numerical implementation: Although the theoretical framework accounts for a temperature-dependent solid heat capacity, in the current THC numerical code implementation, this parameter is maintained as a constant ($C_{ps} = C_{ps0}$). This assumption is made to ensure numerical robustness and simplify the resolution of the highly non-linear energy conservation system at extreme temperatures without significantly affecting the overall thermal diffusion profile.

The thermal conductivity depends on saturation, porosity, and temperature [**baggio1993**]:

$$\lambda = \lambda_d \left(1 + 4 \frac{S_l \, \phi \, \rho_l}{(1-\phi)\, \rho_s}\right) \tag{A.9}$$

with [**harmathy1973**]:

$$\lambda_d = \lambda_{d0} \left[1 + A_\lambda \left(T - T_0\right)\right] \tag{A.10}$$

As drying progresses ($S_l \to 0$), both ρC_p and λ decrease, which accelerates the penetration of the thermal front into the material. These dependencies highlight the influence of the hydric state on the thermal field, which is further discussed in Appendix A.5.1.

Discretization

Using the FE discretization, the temperature and temperature gradient in the spatial domain are expressed in matrix form as:

$$T(x) = [N(x)]\{T\}, \quad \overrightarrow{\text{grad}}(T) = [B(x)]\{T\} \tag{A.11}$$

From the weak formulation of the heat equation combined with the spatial discretization, we obtain the general matrix equation:

$$[C]\{\dot{T}\} + [K]\{T\} = \{F\} \tag{A.12}$$

Where the matrices are defined as follows:

- **Capacity matrix:**

$$[C] = \int_V \rho c_p [N]^T [N] dV \quad (J/K) \tag{A.13}$$

- **Conductivity matrix:**

$$[K] = \int_V [B]^T [\lambda] [B] dV + \int_{\partial V^\varphi} h [N]^T [N] dS \quad (W/K) \tag{A.14}$$

- **Equivalent nodal flux vectors:**

$$\{F\} = \int_V [N]^T q dV + \int_{\partial V^\varphi} [N]^T \left(\varphi_{imp} + hT_f + \varepsilon\sigma(T_\infty^4 - T^4)\right) dS \quad (W) \tag{A.15}$$

For steady-state thermal analysis, the temperature is time-independent ($\{\dot{T}\} = 0$) (assuming constant material properties and neglecting radiation):

$$[K]\{T\} = \{F\} \tag{A.16}$$

By solving this linear system, we can determine the unknown temperatures $\{T\}$ at the nodes across the entire global mesh.

A.1.3 Thermal gradient

Due to the low thermal conductivity of concrete, fire exposure produces steep temperature gradients near the heated surface. The hot surface layer tends to expand while the cooler interior restrains this deformation, generating compressive stresses at the surface and tensile stresses in the interior. Since concrete is inherently weak in tension, this stress state can initiate cracking parallel to the heated face. The role of thermal gradient in the spalling process is discussed further in Appendix A.8 of Chapter 2.

A.2 Hydric behavior

[REVIEW]
When concrete is exposed to high temperatures, the behavior of water within the porous network plays a central role in the thermo-hydro-mechanical response of the material. This section presents the classification of water in concrete, the phase-change mechanisms, the governing equation for moisture transport, and the constitutive laws required to close the system.

A.2.1 Types of water in concrete

Concrete is a multiphase porous material whose voids can be filled with water, air and other fluids. The water present in the cement paste can be broadly classified into two categories [6]

- **Pore water (free/evaporable water):** This water evaporates when the thermodynamic conditions are met (*i.e.*, at $T = 100\,°\mathrm{C}$ and atmospheric pressure) and exists in three main forms:
  - *Capillary and inter-hydrate water:* fills the capillary pores (size $\sim$ 100s of nm) and inter-hydrate pores(size $\sim$ 10s of nm), corresponding to a saturation $S_{ssp} < S_l < 1$.

  - *Physically adsorbed water (gel water):* retained by the surface of the solid matrix due to Van der Waals forces, confined in gel pores (size $\sim$ a few nm).
  - *Interlayer water:* confined in $\sim$ 1 nm space between the C-S-H layers.

- **Chemically bound water (non-evaporable):** This water is combined with anhydrous cement during hydration and forms part of the crystalline structure of hydration products (e.g., C-S-H, portlandite). It can only be released through thermal decomposition (dehydration) at elevated temperatures.

It is important to note that the range of pore sizes is a continuous one; the dimensional boundaries between capillary, inter-hydrate and gel pores are approximations to inherently blurred boundaries.

A.2.2 Evaporation and dehydration

Two distinct phase-change mechanisms govern the release of water in heated concrete:

- **Evaporation** is the transition of free liquid water into vapor. It occurs progressively as temperature increases, becoming significant above approximately 100 °C at atmospheric pressure. The evaporation rate $\dot{m}_{vap}$ is determined from the mass conservation of liquid water (see Appendix A.2.3).

- **Dehydration** is the thermally-driven decomposition of hydration products, releasing chemically bound water as vapor. The main reactions occur at characteristic temperature ranges [6]:
  - Up to 300 °C: progressive decomposition of C-S-H phases and ettringite;
  - 400–500 °C: dehydroxylation of portlandite ($Ca(OH)_2 \rightarrow CaO + H_2O$);
  - $\sim$ 570 °C: α–β quartz transformation;
  - Above 600 °C: decarbonation of calcareous aggregates ($CaCO_3 \rightarrow CaO + CO_2$).

Both evaporation and dehydration act as vapor sources that increase gas pressure within the pore network. They also consume latent energy, affecting the temperature field (see Appendix A.5).

Dehydration kinetics

The cumulative dehydrated water mass is described by the following phenomenological relationship [**pesavento2000**]:

$$\Delta m_{dehyd} = f_s \cdot f_m \cdot f_c \cdot f(T) \tag{A.17}$$

where f_s is a stoichiometric factor, f_m accounts for the concrete's age, f_c is the cement mass per unit volume, and $f(T)$ is a temperature-dependent function:

$$f(T) = \begin{cases} 0 & \text{if } T < 105\,^\circ\text{C} \\ \left[1 + \sin\!\left(\frac{\pi}{2}\,(1 - 2\exp(-0.004\,(T - 105)))\right)\right] & \text{if } T \geq 105\,^\circ\text{C} \end{cases} \tag{A.18}$$

Enthalpy of evaporation

The enthalpy of evaporation is temperature-dependent and is approximated by the Watson formula [**reid1987**]:

$$\Delta H_{vap}(T) = \Delta H_{vap,0} \left(\frac{T_{crit} - T}{T_{crit} - T_0}\right)^{0.38} \tag{A.19}$$

where $T_{crit} = 647$ K is the critical temperature of water and $\Delta H_{vap,0} = 2.672 \times 10^5$ J/kg.

A.2.3 Moisture transport equation

Transport mechanisms

The transport of moisture in heated concrete is governed by two mechanisms:

- **Advective flow (Darcy's law):** The bulk flow of liquid water and gas through the porous network is driven by pressure gradients:

$$\mathbf{v}_{l-s} = -\frac{K\,k_{rl}}{\mu_l}\,(\nabla p_g - \nabla p_c) \tag{A.20}$$

$$\mathbf{v}_{g-s} = -\frac{K\,k_{rg}}{\mu_g}\nabla p_g \tag{A.21}$$

  where K is the intrinsic permeability, k_{rl} and k_{rg} are the relative permeabilities of liquid and gas, μ_l and μ_g are the dynamic viscosities, and $p_l = p_g - p_c$ is the liquid pressure.

- **Diffusive flow (Fick's law):** Water vapor diffuses within the gas mixture according to:

$$\mathbf{j}_v = -\rho_g\, D\, \nabla \left(\frac{\rho_v}{\rho_g}\right) \tag{A.22}$$

  where D is the effective diffusivity accounting for the tortuous porous network.

Mass conservation equations

The mathematical framework for moisture transport in heated concrete is based on the conservation of mass for each phase. The general form of a mass conservation equation for a phase π is [6]:

$$\frac{\partial}{\partial t}(\text{density}) + \nabla \cdot (\text{flux}) = \text{source} \tag{A.23}$$

Concrete at high temperature is treated as a multiphase porous medium consisting of a solid skeleton (s), liquid water (l), water vapor (v), and dry air (a). Applying Equation (A.23) to each phase yields [6]:

- **Solid skeleton:**
$$\frac{\partial m_s}{\partial t} = \dot{m}_{dehyd} \tag{A.24}$$
- **Liquid water:**
$$\frac{\partial m_l}{\partial t} + \nabla \cdot (m_l \, \mathbf{v}_{l-s}) = -\dot{m}_{vap} - \dot{m}_{dehyd} \tag{A.25}$$
- **Water vapor:**
$$\frac{\partial m_v}{\partial t} + \nabla \cdot (m_v \, \mathbf{v}_{g-s}) + \nabla \cdot (m_v \, \mathbf{v}_{v-g}) = \dot{m}_{vap} \tag{A.26}$$
- **Dry air:**
$$\frac{\partial m_a}{\partial t} + \nabla \cdot (m_a \, \mathbf{v}_{g-s}) + \nabla \cdot (m_a \, \mathbf{v}_{a-g}) = 0 \tag{A.27}$$

where $\mathbf{v}_{\pi-s}$ is the velocity of phase π ($\pi = l, g$) relative to the solid skeleton, and $\mathbf{v}_{\pi-g}$ is the velocity of species π ($\pi = v, a$) relative to the gas mixture. The source term $\dot{m}_{vap}$ represents the evaporation rate ($\dot{m}_{vap} > 0$ for evaporation, < 0 for condensation), and $\dot{m}_{dehyd}$ is the dehydration rate ($\dot{m}_{dehyd} < 0$, treated as a mass loss of the solid skeleton and a mass source for liquid water).

The mass of each phase per unit volume of porous medium is [6]:

$$m_s = (1 - \phi)\, \rho_s \tag{A.28}$$
$$m_l = \rho_l \, S_l \, \phi \tag{A.29}$$
$$m_v = \rho_v \, (1 - S_l)\, \phi \tag{A.30}$$
$$m_a = \rho_a \, (1 - S_l)\, \phi \tag{A.31}$$

where ϕ is the porosity and S_l is the liquid saturation degree.

By combining the mass conservation of liquid water and water vapor, the evaporation source term $\dot{m}_{vap}$ is eliminated, yielding the **water species conservation equation**:

$$\begin{aligned} &S_l\rho_l\frac{\partial\phi}{\partial t} + S_l\phi\frac{\partial\rho_l}{\partial t} + \rho_l\phi\frac{\partial S_l}{\partial t} + (1-S_l)\phi\frac{\partial\rho_v}{\partial t} - \rho_v\phi\frac{\partial S_l}{\partial t} + (1-S_l)\rho_v\frac{\partial\phi}{\partial t} \\ &+\nabla\cdot\left(-K\frac{\rho_l k_{rl}}{\mu_l}\nabla p_l\right) + \nabla\cdot\left(-K\frac{\rho_v k_{rg}}{\mu_g}\nabla p_g - D\rho_g\frac{M_v M_a}{M_g^2}\nabla\frac{p_v}{p_g}\right) = -\dot{m}_{dehyd} \end{aligned} \tag{A.32}$$

This equation has three primary unknowns: the temperature T, the capillary pressure p_c, and the gas pressure p_g. Similarly, the **dry air conservation equation** reads:

$$(1-S_l)\phi\frac{\partial\rho_a}{\partial t} - \rho_a\phi\frac{\partial S_l}{\partial t} + (1-S_l)\rho_a\frac{\partial\phi}{\partial t} + \nabla\cdot\left(-K\frac{\rho_a k_{rg}}{\mu_g}\nabla p_g - D\rho_g\frac{M_v M_a}{M_g^2}\nabla\frac{p_a}{p_g}\right) = 0 \tag{A.33}$$

Constitutive relations for the gas phase

The gaseous phase (dry air + water vapor) is governed by the following relations [6]:

- **Ideal gas law:**

$$\rho_\pi = \frac{p_\pi M_\pi}{RT}, \quad \pi = a, v \tag{A.34}$$

  where M_π is the molar mass and R is the universal gas constant.

- **Dalton's law:**

$$p_g = p_a + p_v \tag{A.35}$$

- **Kelvin equation** (thermodynamic equilibrium between liquid and vapor):

$$p_v = p_{vs}(T)\exp\left(\frac{M_v}{\rho_l RT}(p_g - p_c - p_{vs})\right) \tag{A.36}$$

  where $p_{vs}(T)$ is the saturation vapor pressure, approximated by the Hyland-Wexler relationship [**ashrae2001**]:

$$p_{vs}(T) = \exp\left[\frac{C_1}{T} + C_2 + C_3 T + C_4 T^2 + C_5 T^3 + C_6\ln(T)\right] \tag{A.37}$$

Sorption isotherms

The relationship between capillary pressure and liquid saturation is described by the Van Genuchten model [**vangenuchten1980**, **baroghel1999**]:

$$p_c = a\left(S_l^{-b} - 1\right)^{1-1/b} \tag{A.38}$$

where a and b are material parameters determined experimentally. Representative values at ambient temperature are given in Table A.1.

Table A.1: Van Genuchten parameters at ambient temperature

	NSC	HPC	NSM	HPM
a [MPa]	18.62	46.94	37.55	96.28
b [-]	2.275	2.060	2.168	1.954

At elevated temperatures, the parameter a is modified to account for changes in surface tension and pore structure [**pesavento2000**]:

$$S_l = \left(\left(\frac{E(T)}{a(T)} \, p_c \right)^{b/(b-1)} + 1 \right)^{-1/b} \tag{A.39}$$

where $E(T)$ accounts for the surface tension evolution and $a(T)$ accounts for microstructure changes above 100 °C.

Intrinsic permeability

Permeability is a key parameter governing moisture transport. For a thermo-hydral problem, the following relationship is adopted [**gawin1999**]:

$$K = K_0 \cdot 10^{A_T(T-T_0)} \left(\frac{p_g}{p_{atm}} \right)^{A_p} \tag{A.40}$$

where K_0 is the intrinsic permeability at reference conditions, $A_T = 0.005$ and $A_p = 0.368$.

When mechanical damage is considered, the permeability increases significantly [**gawin2002**]:

$$K = K_0 \cdot 10^{A_T(T-T_0)} \left(\frac{p_g}{p_{atm}} \right)^{A_p} \cdot 10^{A_D \, \mathcal{D}} \tag{A.41}$$

where $\mathcal{D}$ is the damage parameter and A_D is a material constant. This damage-permeability coupling is a key mechanism in the THM model (see Appendix A.5).

Relative permeability

The relative permeabilities of liquid and gas are described by the Van Genuchten relationships [**vangenuchten1980**]:

$$k_{rl} = \sqrt{S_l} \left(1 - \left(1 - S_l^{1/A_S} \right)^{A_S} \right)^2 \tag{A.42}$$

$$k_{rg} = \sqrt{1 - S_l} \left(1 - S_l^{1/A_S} \right)^{2A_S} \tag{A.43}$$

where $A_S = 0.6$.

Effective diffusivity

The diffusivity of vapor in air at a given temperature and pressure is [**perre1990**]:

$$D_{va}(T, p_g) = D_{v0} \left(\frac{T}{T_0} \right)^{B_v} \frac{p_{g0}}{p_g} \tag{A.44}$$

where $D_{v0} = 2.58 \times 10^{-5}$ m^2/s and $B_v = 1.667$. The effective diffusivity in the porous medium accounts for saturation and tortuosity:

$$D = (1 - S_l)^{A_v} \, \phi \, \tau \, D_{va}(T, p_g) \tag{A.45}$$

with the tortuosity:

$$\tau(\phi, S_l) = \phi^{1/3}(1 - S_l)^{7/3} \tag{A.46}$$

Fluid viscosities

The dynamic viscosities are temperature-dependent [**ashrae2001**]. The gas viscosity is a concentration-weighted average:

$$\mu_g = \frac{p_v}{p_g}\mu_v + \frac{p_a}{p_g}\mu_a \tag{A.47}$$

where μ_v and μ_a are linear functions of temperature. The liquid water viscosity decreases strongly with temperature [**thomas1995**]:

$$\mu_l = \mu_{l0} \exp\left(\frac{-\alpha_l (T - T_0)}{T} \right) \tag{A.48}$$

with $\mu_{l0} = 1.002 \times 10^{-3}$ Pa·s and $\alpha_l = 0.6612$.

Porosity evolution

The porosity increases with temperature due to microstructural changes [**schneider1989**]:

$$\phi = \phi_0 + A_\phi \, (T - T_0) \tag{A.49}$$

where ϕ_0 is the initial porosity and A_ϕ is a material constant.

A.2.4 Pore pressure build-up

When concrete is heated, evaporation and dehydration produce vapor that raises the pore pressure. Part of this vapor migrates toward the cooler interior, where it recondenses and forms a quasi-saturated zone — the *moisture clog*. This layer blocks further gas transport, causing significant pressure build-up behind the drying front. In dense concretes such as HPC, the low intrinsic permeability K_0 further limits gas evacuation.

When the pore pressure exceeds the tensile strength, it can trigger explosive spalling (Appendix A.8).

A.3 Chemical behavior and Microstructural evolution

A.3.1 Powers' stoichiometric model and water content

The initial hydro-chemical state of the concrete, including the maximum water content available for hydration and dehydration, is derived from its mix-design proportions $(c, w/c, s/c)$ using Powers' stoichiometric model [14]. The theoretical maximum hydrated cement fraction ξ_∞ is computed by assuming no capillary water remains at the end of the reaction ($V_{cw} = 0$), yielding:

$$\xi_\infty = \min\left\{\frac{p}{k[1.32 + 1.57(s/c)](1-p)},\ 1\right\} \tag{A.50}$$

where p is the initial volume fraction of water and k is a density-dependent mix parameter. The maximum chemically bound water at full hydration is then internally computed as:

$$m_{hyd}^{\infty} = 0.228\, c\, \xi_\infty \tag{A.51}$$

This stoichiometric quantity is fundamental, as it dictates the total amount of water that can be released during high-temperature dehydration. Furthermore, the latent heat of hydration L_{hyd} (often denoted as Q_{lat} in the numerical code), which acts as the heat source in the energy conservation equation, is directly linked to this bound water by:

$$L_{hyd} = 0.228\, c\, \xi_\infty H_{hyd} \tag{A.52}$$

where H_{hyd} is the specific enthalpy of hydration.

A.3.2 Chemical kinetics and Affinity

The evolution of the hydration degree Γ is governed by an Arrhenius-type chemo-kinetic equation:

$$\frac{\mathrm{d}\Gamma}{\mathrm{d}t} = \mathcal{A}(\Gamma) \cdot \beta_{\mathrm{RH}} \cdot \exp\left(-\frac{E_a}{RT}\right) \tag{A.53}$$

where E_a/R (input parameter EAR) represents the activation energy scaled by the universal gas constant, dictating the sensitivity of the reaction rate to temperature. β_{RH} accounts for the deceleration of hydration at low moisture levels.

The macroscopic chemical affinity $\mathcal{A}(\Gamma)$ drives the reaction speed depending on the current hydration state [14]:

$$\mathcal{A}(\Gamma) = \frac{A_i + (A_p - A_i)\sin\left[\frac{\pi}{2}\left(1 - \left\langle \frac{\Gamma_p - \Gamma}{\Gamma_p} \right\rangle_+\right)\right]}{\left[1 + \zeta \left\langle \frac{\Gamma - \Gamma_p}{1 - \Gamma_p} \right\rangle_+\right]} - \left(\frac{A_p}{1+\zeta}\right)\left\langle \frac{\Gamma - \Gamma_p}{1 - \Gamma_p} \right\rangle_+ \tag{A.54}$$

where $\langle \cdot \rangle_+$ is the positive part operator. The parameters A_i, A_p regulate the initial and peak affinity, Γ_p is the hydration degree at the peak, and ζ controls the spatial hindrance as hydrates precipitate.

A.3.3 Porosity evolution law

The porosity of the cement paste is dynamically altered by the hydration and dehydration processes. Based on Powers' volumetric model [14], the rate of change of porosity is directly proportional to the effective hydration degree:

$$\frac{\partial \phi}{\partial t} = -a_\phi \Omega \frac{\partial \tilde{\Gamma}}{\partial t} \tag{A.55}$$

where a_ϕ is a material parameter dictating the volume of hydrates formed or destroyed, and Ω is the volume fraction of the cement paste in the concrete mix. Integrating this rate yields the explicit dependence of porosity on the chemical state:

$$\phi(\tilde{\Gamma}) = \phi_0 + a_\phi \Omega (1 - \tilde{\Gamma}) \tag{A.56}$$

where ϕ_0 is the reference porosity at the fully hydrated baseline state ($\tilde{\Gamma} = 1$).

A.3.4 Permeability evolution law

The intrinsic permeability is heavily dependent on the degradation of the solid matrix. In the THC framework, it is explicitly coupled to the effective hydration degree $\tilde{\Gamma}$ [14]:

$$K(\tilde{\Gamma}) = K_0 \cdot 10^{A_\Gamma (1-\tilde{\Gamma})^{B_K}} \tag{A.57}$$

where K_0 is the initial intrinsic permeability, A_Γ is a coefficient defining the maximum increase in permeability upon total dehydration, and B_K modifies the shape of the permeability gain.

A.4 Mechanical behavior

[REVIEW]

The mechanical behaviour is modelled by coupling linear elasticity with continuum damage mechanics (Mazars model), described in the following subsections.
The TH and mechanical problems are solved with a staggered (sequential) solution strategy: the TH subsystem is solved first to obtain p_g, p_c, T, which are then passed as known loading to the mechanical step.
The specific coupling method between the TH and mechanical solvers has not yet been finalized and will be detailed and modified in Chapter 2 and Chapter 3 as the study progresses.

A.4.1 Linear elastic model

In the present work, concrete is modeled as a linear elastic material. The effective stress tensor $\tilde{\boldsymbol{\sigma}}$, defined as the stress acting on the undamaged material, is related to the elastic strain tensor $\boldsymbol{\varepsilon}_e$ through the isotropic Hooke's law:

$$\boldsymbol{\sigma} = \mathbb{C} : \boldsymbol{\varepsilon}_e \tag{A.58}$$

where $\mathbf{C}$ is the constant fourth-order elastic stiffness tensor, defined by the Young's modulus E and Poisson's ratio ν.

The total strain is decomposed into mechanical and free (stress-free) components, as detailed in Appendix A.5.2. Since creep is neglected in this work, the elastic strain reduces to:

$$\boldsymbol{\varepsilon}_e = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_{th} - \boldsymbol{\varepsilon}_{sh} \tag{A.59}$$

which enters directly into Hooke's law (Equation (A.58)).

Linear finite element formulation

The spatial discretization of the momentum conservation equation (Equation (A.87)) is carried out using the standard Galerkin finite element method [4].

The continuous displacement field $\mathbf{u}$ is approximated by nodal displacements $\{\mathbf{U}\}$ using shape functions $\mathbf{N}$, and the strain is obtained via the compatibility matrix $\mathbf{B}$ such that $\boldsymbol{\varepsilon} = \mathbf{B}\{\mathbf{U}\}$.

Applying the principle of virtual work, the global system takes the form:

$$[\mathbf{K}_M]\{\mathbf{U}\} = \{\mathbf{F}\} \tag{A.60}$$

where $[\mathbf{K}_M] = \int_V \mathbf{B}^T \mathbb{C}\, \mathbf{B}\, dV$ is the elastic stiffness matrix. Its generalisation to account for damage is given in Equation (A.70).

The load vector $\{\mathbf{F}\}$ receives contributions from external forces and from the thermo-hydric fields: (Appendix A.5.1):

$$\{\mathbf{F}\} = \{\mathbf{f}_{ext}\} + \underbrace{\int_V \mathbf{B}^T \mathbb{C}\, \boldsymbol{\varepsilon}^{th}\, dV + \int_V \mathbf{B}^T \mathbb{C}\, \boldsymbol{\varepsilon}^{sh}\, dV}_{\text{strain decomposition (Equation (A.78))}} + \underbrace{\int_V \mathbf{B}^T\, \alpha\, p_s\, \mathbf{I}\, dV}_{\text{effective stress (Equation (A.84))}} \tag{A.61}$$

where $\{\mathbf{f}_{ext}\}$ groups the classical external forces. All thermo-hydric terms are computed from the TH solution at each time step and enter as prescribed loading, detailed in Appendix A.5.1.

In Cast3M, this system is assembled using the `RIGI` and `RESO` operators for the linear part. When damage is included, the non-linear iterative solution is handled by `PASAPAS` (Equation (A.71)).

A.4.2 Mazars damage model

Concrete cracks when tensile strains become too large. In this work, this behaviour is represented with the Mazars damage model [**mazars1986**, 10].

Damage parameter and equivalent strain

The model introduces a single scalar damage variable $\mathcal{D} \in [0, 1]$ [**mazars1986**]. The effective stress in the damaged material is:

$$\boldsymbol{\sigma}' = (1 - \mathcal{D})\, \mathbb{C}\, \boldsymbol{\varepsilon}_{el} \tag{A.62}$$

When $\mathcal{D} = 0$, this reduces to Hooke's law (Equation (A.58)). The damage variable is driven by the equivalent strain, computed from the positive principal elastic strains:

> **[REVIEW]**
>
> Because concrete cracking is primarily governed by tensile extensions, Mazars introduced an *equivalent strain* (ε_{eq}) to translate any 3D multiaxial strain state into a scalar uniaxial indicator. In the case of a thermo-mechanical loading, it is important to note that only the elastic strain tensor $\boldsymbol{\varepsilon}_e$ contributes to the evolution of damage.

The equivalent strain is calculated solely from the positive principal elastic strains ε_i:

$$\varepsilon_{eq} = \sqrt{\langle \boldsymbol{\varepsilon}_e \rangle_+ : \langle \boldsymbol{\varepsilon}_e \rangle_+} = \sqrt{\sum_{i=1}^{3} \langle \varepsilon_i \rangle_+^2} \tag{A.63}$$

where the Macaulay brackets $\langle x \rangle_+$ denote the positive part of x (i.e., $\langle x \rangle_+ = x$ if $x \geq 0$, and 0 if $x < 0$).

Damage initiates when the equivalent strain exceeds the initial damage threshold, denoted as ε_{d0}.

$$\varepsilon_{d0} = \frac{f_t}{E}. \tag{A.64}$$

At the damage onset ($\varepsilon_{eq} = \varepsilon_{d0}$), the damage variable is zero:

$$\mathcal{D} = 0 \qquad \text{for} \qquad \varepsilon_{eq} \leq \varepsilon_{d0}. \tag{A.65}$$

For uniaxial tension, damage onset corresponds to:

$$\sigma_t = f_t. \tag{A.66}$$

In this simplified interpretation, f_t denotes the uniaxial tensile strength of the concrete and E is Young's modulus.

Damage evolution laws for tension and compression

To accurately represent concrete behavior, the total damage D is formulated as a combination of two independent damage variables: D_t for tension and D_c for compression.

When the damage threshold is exceeded ($\varepsilon_{eq} > \varepsilon_{d0}$), the evolution of these two modes is explicitly governed by exponential laws:

$$\mathcal{D}_t = 1 - \frac{(1 - A_t)\varepsilon_{d0}}{\varepsilon_{eq}} - A_t \exp\left[-B_t(\varepsilon_{eq} - \varepsilon_{d0})\right] \tag{A.67}$$

$$\mathcal{D}_c = 1 - \frac{(1 - A_c)\varepsilon_{d0}}{\varepsilon_{eq}} - A_c \exp\left[-B_c(\varepsilon_{eq} - \varepsilon_{d0})\right] \tag{A.68}$$

where A_t, B_t, A_c, and B_c are material parameters identified from uniaxial tensile and compressive tests. These parameters modulate the shape of the post-peak softening curves: parameters A define the residual asymptotic stress level, while parameters B control the slope of the stress drop.

The global macroscopic damage $\mathcal{D}$ is then calculated using a linear combination with weighting coefficients α_t and α_c:

$$\mathcal{D} = \alpha_t^{\beta}\mathcal{D}_t + \alpha_c^{\beta}\mathcal{D}_c \tag{A.69}$$

The coefficients α_t and α_c depend on the principal stress state, evaluating the respective contributions of tensile and compressive stresses. In pure tension, $\alpha_t = 1$ and $\alpha_c = 0$; conversely, in pure compression, $\alpha_c = 1$ and $\alpha_t = 0$. The parameter β (typically fixed at 1.06) is a shear-correction coefficient introduced to improve the structural response under shear loadings.

Non-linear equilibrium and iterative solution

In the general case with damage, the stiffness matrix reads:

$$[\mathbf{K}_M] = \int_V \mathbf{B}^T \left(1 - \mathcal{D}\right) \mathbb{C}\, \mathbf{B}\, dV \tag{A.70}$$

When $\mathcal{D} = 0$ (intact material), this reduces to the standard linear elastic stiffness matrix $\int_V \mathbf{B}^T \mathbb{C}\, \mathbf{B}\, dV$.

Due to damage evolution, the constitutive relation becomes non-linear and Equation (A.60) cannot be solved in a single step. The equilibrium is enforced iteratively using a Newton-Raphson-type algorithm [4]. At each iteration i, the stiffness matrix (Equation (A.70)) is evaluated with the current damage $\mathcal{D}^i$, and the correction is obtained from:

$$[\mathbf{K}_M]^i \{\delta\mathbf{U}\}^{i+1} = \{\mathbf{F}\} - \{\mathbf{R}\}^i \tag{A.71}$$

where:

- $\{\mathbf{R}\}^i = \int_V \mathbf{B}^T \boldsymbol{\sigma}^i\, dV$ is the internal force vector, with $\boldsymbol{\sigma}^i = (1 - \mathcal{D}^i)\, \mathbb{C}\, \boldsymbol{\varepsilon}_{el}^i$;
- $\{\mathbf{F}\}$ is the external load vector including all thermo-hydric contributions (Equation (A.61)).

The displacements and damage are updated until convergence: $\|\{\mathbf{F}\} - \{\mathbf{R}\}^i\| < \epsilon\, F_{ref}$. In Cast3M, this iterative procedure is handled by the `PASAPAS` operator.

A.4.3 Mesh dependency and fracture energy regularization

A well-known drawback of local softening models such as Mazars' is the pathological mesh dependency: softening causes strain localization in a band whose width depends on the element size, leading to vanishing energy dissipation upon mesh refinement [**bazant1976**].

Two main families of regularization exist in the literature:

- **Nonlocal averaging:** the local equivalent strain ε_{eq} is replaced by a weighted spatial average $\bar{\varepsilon}_{eq}$ over a characteristic volume controlled by an internal length l_c [**pijaudier1987**, **giry2011**]. This is the approach adopted by [6].
- **Fracture energy regularization (Hillerborg):** the softening law parameters are adjusted element-by-element so that each element dissipates the same fracture energy G_f per unit crack area, regardless of its size [**hillerborg1976**]. This is the approach adopted in the present work, as it is simpler to implement in Cast3M and does not require additional nonlocal operators.

Principle of fracture energy regularization

Physically, the fracture energy G_f [**hillerborg1976**] represents the energy per unit area required to propagate a crack. In the continuum damage framework, cracking is smeared over the element volume. A characteristic length L_e bridges the two descriptions: in Cast3M (using `@LONGEF`), $L_e = V^{1/D}$ where D is the spatial dimension.

The total dissipated energy per unit crack area is:

$$G_f = G_{elastic} + G_{softening} \tag{A.72}$$

Elastic energy ($G_{elastic}$)

The elastic energy density up to damage onset is:

$$g_{elastic} = \frac{1}{2} f_t \, \varepsilon_{d0} = \frac{f_t^2}{2E_0} \tag{A.73}$$

so that $G_{elastic} = g_{elastic} \, L_e = \frac{f_t^2}{2E_0} \, L_e$, and the energy available for softening is:

$$G_{softening} = G_f - \frac{f_t^2}{2E_0} \, L_e \tag{A.74}$$

Regularization of B_t

Assuming $A_t \approx 1$ (brittle tension), the post-peak stress simplifies to $\sigma \approx f_t \exp[-B_t(\varepsilon - \varepsilon_{d0})]$, whose integral gives $g_{softening} = f_t/B_t$. Equating $G_{softening} = g_{softening} \cdot L_e$ yields the modified parameter:

$$B_t^{mod} = \frac{f_t \, L_e}{G_f - \frac{f_t^2}{2E_0} \, L_e} \tag{A.75}$$

This ensures that every element dissipates exactly G_f regardless of mesh size, making the global response objective with respect to spatial discretization.

A.5 Coupling mechanisms

The thermal, hydric, and mechanical behaviors presented in the previous sections are not independent: they interact through a network of coupling mechanisms. This section identifies and formulates the two main coupling paths that govern the response of concrete exposed to fire.

A.5.1 Thermo-Hydric (TH) coupling mechanisms

The thermal and hydric fields are coupled in both directions: temperature drives phase changes and modifies transport properties, while fluid flow and phase changes affect the energy balance.

Temperature as a driver of hydric phenomena

Temperature enters the hydric conservation equations (Equations (A.32) and (A.33)) through two pathways:

- **Mass source terms:** evaporation and dehydration (Appendix A.2.2) are thermally activated processes that inject vapor into the pore network, appearing as source terms $\dot{m}_{vap}$ and $\dot{m}_{dehyd}$ in the mass balance equations;
- **Transport coefficients:** most constitutive parameters are temperature-dependent — permeability $K(T)$, porosity $\phi(T)$, fluid viscosities $\mu(T)$, saturating vapor pressure $p_{vs}(T)$, and diffusivity $D(T)$ — so the coefficients of the mass conservation equations evolve as the thermal field changes.

Energy conservation equation

The energy conservation equation governs the temperature field within the porous medium. Similar to the heat equation in Appendix A.1.2, it includes a storage term, a conductive flux, and additionally accounts for advective heat transport and phase-change energy terms [6]:

$$\rho C_p \frac{\partial T}{\partial t} + \underbrace{(m_l C_l \mathbf{v}_{l-s} + m_g C_g \mathbf{v}_{g-s}) \cdot \nabla T}_{\text{advection}} + \nabla \cdot (-\lambda \cdot \nabla T) = \underbrace{-H_{vap}\, \dot{m}_{vap} + H_{dehyd}\, \dot{m}_{dehyd}}_{\text{phase-change energy}} \tag{A.76}$$

Compared to the classical heat equation Equation (A.5) $\rho C_p \frac{\partial T}{\partial t} - \nabla \cdot (\lambda \nabla T) = 0$, two additional contributions appear:

- **Advective heat transport:** The bulk flow of liquid water and gas, driven by pressure gradients, transport enthalpy through the material. This term redistributes energy along the flow paths and can be significant when permeability and pressure gradients are large.
- **Phase-change energy (right-hand side):** Concrete does not self-generate heat; these terms represent energy consumed by evaporation ($-H_{vap}\, \dot{m}_{vap}$, heat sink since $\dot{m}_{vap} > 0$) and dehydration ($+H_{dehyd}\, \dot{m}_{dehyd}$, also a heat sink since $\dot{m}_{dehyd} < 0$). Both locally slow the temperature rise.

After replacing the mass fluxes by Darcy's law (Equations (A.20) and (A.21)), the energy conservation takes its final coupled form:

$$\begin{aligned} \rho C_p \frac{\partial T}{\partial t} - K \left(C_l \frac{\rho_l k_{rl}}{\mu_l} (\nabla p_g - \nabla p_c) + C_g \frac{\rho_g k_{rg}}{\mu_g} \nabla p_g \right) \cdot \nabla T \\ - \nabla \cdot (\lambda \cdot \nabla T) = -H_{vap}\, \dot{m}_{vap} + H_{dehyd}\, \dot{m}_{dehyd} \end{aligned} \tag{A.77}$$

where the overall heat capacity ρC_p (**??**) and thermal conductivity λ (Equation (A.9)) are defined in Appendix A.1.2.

Thus, the thermal and hydric fields are *strongly coupled*: temperature modifies the source terms and transport coefficients in the moisture equation, while moisture content and fluid flow modify the heat capacity, conductivity, and introduce latent heat and advection terms in the energy equation.

A.5.2 Thermo-hydric input to the mechanical field

The mechanical behaviour of heated concrete is driven by the thermal and hydric fields. The coupling is *one-way*: the converged TH fields (T, p_g, p_c) serve as prescribed loading to the mechanical problem at each time step. The feedback from mechanics to TH through damage-enhanced permeability is discussed in Appendix A.9.

The TH fields enter the mechanical problem through two mechanisms: free strains and pore pressure loading, detailed in the following subsections.

Strain decomposition

When a mechanically loaded concrete element is exposed to high temperature, the total strain is decomposed as a superposition of components of different origins [6]:

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_{\sigma} + \boldsymbol{\varepsilon}_{th} + \boldsymbol{\varepsilon}_{sh} \tag{A.78}$$

where:

- $\boldsymbol{\varepsilon}_{\sigma}$ represents the *mechanical strains* induced by external loading (elastic strain, creep strain, etc.);
- $\boldsymbol{\varepsilon}_{th}$ represents the *thermal dilation strain*, driven by the temperature field;
- $\boldsymbol{\varepsilon}_{sh}$ represents the *capillary shrinkage strain*, driven by the capillary pressure field.

In general, the mechanical strain includes elastic and creep contributions:

$$\boldsymbol{\varepsilon}_{\sigma} = \boldsymbol{\varepsilon}_{el} + \boldsymbol{\varepsilon}_{creep} \tag{A.79}$$

Since creep is neglected in this work, $\boldsymbol{\varepsilon}_{creep} = \mathbf{0}$ and thus $\boldsymbol{\varepsilon}_{\sigma} = \boldsymbol{\varepsilon}_{el}$

Thermal dilation

The thermal dilation strain is isotropic and produces a uniform expansion of the material, which is expressed in incremental form as [6]:

$$d\boldsymbol{\varepsilon}_{th} = \beta_s(T)\, dT\, \mathbf{I} \tag{A.80}$$

where $\mathbf{I}$ is the second-order identity tensor and $\beta_s(T)$ is the thermal dilation coefficient, which for concrete can be approximated as a linear function of temperature:

$$\beta_s(T) = A_{B1}\,(T - T_0) + A_{B0} \tag{A.81}$$

with A_{B0} and A_{B1} being material constants.

Capillary shrinkage

The drying process modifies the pore pressure state, which in turn induces a volumetric deformation of the solid skeleton. The capillary shrinkage strain is expressed as [6]:

$$d\boldsymbol{\varepsilon}_{sh} = \frac{\alpha}{K_T}\,\frac{dp_s}{dt}\,\mathbf{I} \tag{A.82}$$

where α is the Biot coefficient, K_T is the bulk modulus of the drained skeleton, and p_s is the average pore pressure acting on the solid skeleton, defined as:

$$p_s = p_g - S_l\, p_c = S_g\, p_g + S_l\, p_l \tag{A.83}$$

This expression represents the saturation-weighted average of the gas and liquid pressures. Since p_g and p_c are direct outputs of the TH problem, p_s is entirely determined by the TH solution.

Effective stress and pore pressure loading

For a porous medium saturated by multiple fluids, the total stress is decomposed into the part carried by the solid skeleton and the part carried by the pore fluids. Following [**lewis1998**, 6], the effective stress is defined as:

$$\boldsymbol{\sigma}' = \boldsymbol{\sigma}^{Total} + \alpha\, p_s\, \mathbf{I} \tag{A.84}$$

where $\boldsymbol{\sigma}^{Total}$ is the total stress tensor, α is Biot's coefficient, and p_s is the average pore pressure (Equation (A.83)).

The effective stress $\boldsymbol{\sigma}'$ is the stress that governs the mechanical behaviour of the skeleton. In the finite element implementation, this relation is not applied directly; instead, the pore pressure term $\alpha\, p_s$ enters the load vector as an external loading (Equation (A.61)), and the effective stress is recovered from the elastic strain after solving for displacements:

$$\boldsymbol{\sigma}' = (1 - \mathcal{D})\, \mathbb{C}\, \boldsymbol{\varepsilon}_{el} \tag{A.85}$$

where $\boldsymbol{\varepsilon}_{el} = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_{th} - \boldsymbol{\varepsilon}_{sh}$ (Equation (A.59)). This effective stress is the input to the Mazars damage criterion (**??**). In heated concrete, the build-up of gas pressure (Appendix A.2.4) increases p_s in the load vector, which shifts $\boldsymbol{\sigma}'$ toward tension and promotes damage when f_t is exceeded.

A.6 THM interactions and spalling mechanisms

A.6.1 Simultaneous THM behaviors and interactions

When concrete is heated rapidly, it does not merely experience a temperature rise; it undergoes a simultaneous interplay of thermal, hydric, and mechanical (THM) processes that strongly interact with each other. Heat is transmitted into the material essentially by conduction, while phase changes (evaporation of free water and dehydration of chemically bound water) consume energy and locally slow the temperature rise. The generated vapour is transported by pressure-driven Darcy flow and Fickian diffusion. Part of the vapour migrates inward, recondenses in cooler regions, and builds a quasi-saturated layer (the moisture clog) that blocks further gas transport. Mechanically, the concrete skeleton expands due to heat, shrinks due to capillary forces, and is subjected to internal loading from the trapped pressurized gas.

A.6.2 Spalling mechanisms

These coupled THM phenomena are the fundamental drivers of *explosive spalling*. The specific mechanisms contributing to this failure are detailed below:

- **Thermal:** Steep temperature gradients develop throughout the concrete's cross-section, causing differential thermal expansion. The surface layers attempt to expand but are restrained by the cooler interior, generating high compressive stresses at the heated surface and tensile stresses deeper inside.
- **Hydric:** In low-permeability concretes such as high-performance concrete (HPC), the rapid generation of vapour from phase changes cannot escape fast enough. The formation of the moisture clog further seals the porous network, resulting in a dramatic pore pressure build-up behind the drying front.
- **Mechanical:** The combined effects of thermal stresses and pore pressure build-up on the solid skeleton stretch the material beyond its tensile capacity. High temperatures simultaneously degrade the tensile strength $f_t(T)$ of the concrete.

When the combined stress from the thermal gradient $\sigma_{thermal}$ and the pore pressure σ_{pore} exceeds the degraded tensile strength of the concrete, cracks initiate parallel to the heated surface. The sudden release of accumulated elastic energy leads to the violent ejection of concrete fragments, characterizing the explosive spalling phenomenon.

A.7 Thermal and hydric phenomena in heated concrete

When one face of a concrete member is exposed to a rising temperature, a sequence of strongly interacting thermal and hydric processes develops through the thickness. Heat is transmitted into the material essentially by conduction, while phase changes (evaporation, dehydration) consume energy and locally slow the temperature rise. The generated vapour is transported by pressure-driven Darcy flow and Fickian diffusion. Part of the vapour migrates inward, recondenses in cooler regions, and builds a quasi-saturated layer (the moisture clog) that blocks further gas transport, causing a sharp pore-pressure build-up. The full mathematical derivation of these physical mechanisms, including fluxes and constitutive relations, is detailed in Appendix A.

A.8 Spalling mechanisms

Despite extensive experimental work, no single theory has reached consensus; the two most widely accepted mechanisms, and their combined action, are presented below.

A.8.1 Thermal-gradient mechanism

A steep temperature gradient through the thickness produces differential thermal dilation:

1. **Compressive stresses in the heated cover**: the surface layers tend to expand but are restrained by the cooler interior, generating compressive stresses parallel to the heated face.
2. **Tensile stresses in the interior**: the restraint of the expanding surface induces tension in the deeper layers.
3. **Fracture and energy release**: when the tensile stress exceeds the tensile strength f_t, cracks form parallel to the surface; the sudden release of stored elastic energy can drive the explosive ejection of fragments.

High-performance concrete (HPC) is particularly prone to this mechanism because its higher brittleness stores more elastic energy before failure, leading to a more violent release.

A.8.2 Pore-pressure mechanism (moisture clog)

The sequence of events is:

1. **Dry zone**: near the heated face, free water evaporates and bound water is released by dehydration, forming a dry zone.
2. **Vapour migration**: part of the vapour escapes towards the heated face, part is driven inward.
3. **Condensation / moisture clog**: the inward vapour condenses where $T < T_{sat}$, forming a quasi-saturated, nearly impermeable barrier.

4. **Pressure build-up**: behind the clog, p_g rises sharply as evaporation feeds a region from which vapour cannot escape; thermal dilation of liquid water in saturated pores adds to the build-up.

5. **Tensile failure**: when p_g exceeds the tensile strength, cracks form parallel to the surface, potentially leading to explosive spalling.

The governing parameters are the intrinsic permeability K_0 (low permeability hinders vapour evacuation) and the initial moisture content (more water to evaporate), both promoting higher pore pressures.

A.9 THM governing formulation

The state of concrete at high temperature is described by four primary variables [6]:

$$\mathcal{U} = \{p_g,\, p_c,\, T,\, \mathbf{u}\} \tag{A.86}$$

where p_g is the gas pressure, p_c is the capillary pressure, T is the temperature, and $\mathbf{u}$ is the displacement vector.

The substitution of the constitutive laws and transport fluxes (detailed in the preceding sections) into the general mass, energy, and momentum balances yields the final coupled non-linear system. To avoid redundancy, the fully expanded partial differential equations for dry-air conservation, water-species conservation, and energy conservation are presented directly in Section 2.2 of Chapter 2.

By neglecting inertial forces, the momentum conservation reads:

$$\nabla \cdot \boldsymbol{\sigma}^{Total} + \rho\, \mathbf{g} = \mathbf{0} \tag{A.87}$$

where $\mathbf{g}$ is the gravitational acceleration and the total density ρ is:

$$\rho = (1-\phi)\rho_s + \phi\,(S_l\rho_l + (1-S_l)\rho_g) \tag{A.88}$$

The total stress $\boldsymbol{\sigma}^{Total}$ is related to the effective stress via Bishop's relation (Equation (A.84)), which in turn depends on the primary variables p_g and p_c. The discretisation of this equation into the global system $[\mathbf{K}_M]\{\mathbf{U}\} = \{\mathbf{F}\}$ is presented in Appendix A.4.

A.10 Discretization

A.10.1 Spatial discretization

Applying the standard Galerkin finite element method to the three conservation equations (**??????** in Chapter 2) yields the semi-discrete system [6]:

$$\mathbf{C}\,\dot{\mathbf{Y}} + \mathbf{K}\,\mathbf{Y} = \mathbf{f} \tag{A.89}$$

where $\mathbf{Y} = \{\bar{p}_g,\, \bar{p}_c,\, \bar{T}\}$ is the vector of nodal unknowns. The operators $\mathbf{C}$ (capacity), $\mathbf{K}$ (conductivity), and $\mathbf{f}$ (forcing) are 3×3 block matrices solved simultaneously in a **monolithic** manner for the Thermo-Hydric (TH) subsystem:

$$\mathbf{K} = \begin{pmatrix} K_{gg} & K_{gc} & K_{gt} \\ K_{cg} & K_{cc} & K_{ct} \\ K_{tg} & K_{tc} & K_{tt} \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} C_{gg} & C_{gc} & C_{gt} \\ 0 & C_{cc} & C_{ct} \\ 0 & C_{tc} & C_{tt} \end{pmatrix}, \quad \mathbf{f} = \begin{pmatrix} f_g \\ f_c \\ f_t \end{pmatrix} \tag{A.90}$$

The off-diagonal entries K_{ij}, C_{ij} $(i \neq j)$ encode the strong coupling between the thermal and hydric fields. Detailed expressions of each block are given in [6], Appendix A.

The mechanical problem (Equation (A.87)) is solved in a **staggered** manner: at each time step, the converged TH fields $\bar{T}$, $\bar{p}_g$, $\bar{p}_c$ enter as prescribed loading. The discrete equilibrium (Equation (A.60)) reads:

$$[\mathbf{K}_M]\mathbf{U} = \mathbf{F} \tag{A.91}$$

where $\mathbf{K}_M$ (Equation (A.70)) contains the damaged stiffness $(1-\mathcal{D})\,\mathbb{C}$ — the displacement-dependent part that remains on the left-hand side — and $\mathbf{F}$ (Equation (A.61)) collects the TH-prescribed loads ($\boldsymbol{\varepsilon}^{th}$, $\boldsymbol{\varepsilon}^{sh}$, $\alpha\, p_s$) that do not depend on the displacement vector $\mathbf{U}$.

A.10.2 Temporal discretization (The θ-method)

The time integration of the nonlinear semi-discrete system $\mathbf{C}\dot{\mathbf{Y}} + \mathbf{K}\mathbf{Y} = \mathbf{f}$ is carried out using the Wilson-θ method [**lewis1998**]. The governing equations are evaluated at an intermediate time step $t_{n+\theta} = t_n + \theta\,\Delta t$. This yields a fully discrete algebraic system:

$$\tilde{\mathbf{K}}\Delta\mathbf{Y}^{n+1} + \mathbf{K}\mathbf{Y}^{n} = \mathbf{f}^{n+\theta} \tag{A.92}$$

where the effective stiffness matrix $\tilde{\mathbf{K}}$ is defined as:

$$\tilde{\mathbf{K}} = \frac{\mathbf{C}}{\Delta t} + \theta\mathbf{K} \tag{A.93}$$

The relaxation parameter $\theta \in [5]$ dictates the stability and accuracy of the scheme:

- $\theta = 0$: Explicit forward Euler method (conditionally stable).
- $\theta = 0.5$: Crank-Nicolson scheme (unconditionally stable, second-order accurate but prone to numerical oscillations).
- $\theta = 1$: Fully implicit backward Euler method. This is strictly required ($\theta = 1$) for highly nonlinear THC problems at high temperatures to strongly damp numerical oscillations caused by abrupt phase changes and permeability jumps.

Appendix B

Theory of Stochastic analysis

[REVIEW] This chapter is under review

B.1 Random variable (PDF, CDF, moments, CoV)

[REVIEW] This section is currently under development and will be refined as the study progresses.

A **random variable** X is a quantity whose value is not fixed but described by a probability law. Instead of saying "$E = 30$ GPa", one writes $E \sim \mathcal{N}(30,\, 3^2)$ GPa, meaning that the Young's modulus follows a normal (Gaussian) distribution with mean $\mu = 30$ GPa and standard deviation $\sigma = 3$ GPa.

B.1.1 PDF and CDF

The Probability Density Function (**PDF**) $f_X(x)$ describes the relative likelihood of X taking a given value. It is always non-negative and integrates to one over the entire domain.

The Cumulative Distribution Function (**CDF**) $F_X(x) = P(X \leq x)$ gives the probability that X does not exceed a threshold x:

$$F_X(x) \;=\; \int_{-\infty}^{x} f_X(t)\, \mathrm{d}t. \tag{B.1}$$

Link to stochastic numerical methods: The concepts of PDF and CDF are the mathematical foundations for all uncertainty propagation and reliability methods used in this work:

- **Monte Carlo (MC) and Latin Hypercube Sampling (LHS):** Rely on drawing samples from the input PDFs. Specifically, the CDF is the core concept behind the LHS variance-reduction technique (Appendix B.3.2).
- **Kernel Density Estimation (KDE):** Is a non-parametric method used to reconstruct the continuous empirical PDF of the response from discrete simulated samples.
- **Polynomial Chaos Expansion (PCE):** Acts as a surrogate model to rapidly propagate the input PDFs into an output PDF without running thousands of expensive finite element simulations.
- **Gaussian Quadrature (`PROBDENS`):** Analytically reconstructs the response CDF from its statistical moments to compute the failure probability (Appendix B.4.3).

In reliability analysis, the CDF is used directly to evaluate the probability of exceeding a critical value: $P(X > x_{\text{cr}}) = 1 - F_X(x_{\text{cr}})$ [2].

Two distributions are used in this work:

- **Normal** (Gaussian) distribution $\mathcal{N}(\mu, \sigma^2)$:

$$f_X(x) \;=\; \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]. \tag{B.2}$$

- **Lognormal** distribution $\text{LN}(\mu_{\text{ln}}, \sigma_{\text{ln}}^2)$, where $\ln X$ is normally distributed, ensuring $X > 0$:

$$f_X(x) \;=\; \frac{1}{x\,\sigma_{\text{ln}}\sqrt{2\pi}} \exp\left[-\frac{(\ln x - \mu_{\text{ln}})^2}{2\sigma_{\text{ln}}^2}\right], \qquad x > 0. \tag{B.3}$$

The lognormal distribution is particularly relevant for material properties such as the Young modulus or permeability, which must remain strictly positive [7, 16, 1].

B.1.2 Statistical moments

The distribution of X can be summarised by its **statistical moments** [2], which quantify location, spread, asymmetry, and tail heaviness. In this work, the first two moments are prescribed as *input* to the stochastic analysis, while the higher moments are *computed as output* by the quadrature method (Appendix B.4).

Input moments. The **mean** μ_X and **variance** σ_X^2 (or equivalently the standard deviation $\sigma_X = \sqrt{\mathrm{Var}[X]}$) define the centre and spread of the distribution:

$$\mu_X = \int_{-\infty}^{+\infty} x\, f_X(x)\,\mathrm{d}x, \qquad \sigma_X^2 = \int_{-\infty}^{+\infty} (x-\mu_X)^2\, f_X(x)\,\mathrm{d}x. \tag{B.4}$$

Output moments. The **skewness** δ_3 and **kurtosis** δ_4 measure the asymmetry and tail heaviness of the response distribution:

$$\delta_3 = \frac{E[(X-\mu_X)^3]}{\sigma_X^3}, \qquad \delta_4 = \frac{E[(X-\mu_X)^4]}{\sigma_X^4}. \tag{B.5}$$

These four moments play a central role in the Gaussian Quadrature method (Appendix B.4), where the response PDF is reconstructed from the computed moments using Winterstein or Pearson approximations [15].

B.1.3 Coefficient of Variation (CoV)

The coefficient of variation quantifies the relative dispersion of X:

$$\mathrm{CoV}_X = \frac{\sigma_X}{\mu_X}. \tag{B.6}$$

It is a dimensionless measure that allows comparison of variability across parameters with different units and magnitudes. For concrete, typical values are $\mathrm{CoV}_E \approx 0.10$–$0.20$ for the Young's modulus [7] and $\mathrm{CoV}_K \approx 0.15$–$0.30$ for permeability [1].

B.2 Random fields

[REVIEW] This section is currently under development and will be refined as the study progresses.

A random variable assigns a *single* uncertain value to a material property for the entire structure. In reality, concrete properties vary spatially due to heterogeneous microstructure, local variations in the water-to-cement ratio, and non-uniform compaction during casting [7]. A **random field** captures this spatial variability: it assigns an uncertain value to *every point* $\mathbf{x}$ in the domain, not just one value for the whole structure.

Formally, a random field is a collection of random variables indexed by position [16]:

$$H = \{H(\mathbf{x}, \omega) : \mathbf{x} \in \mathcal{D},\ \omega \in \Omega\}, \tag{B.7}$$

where ω denotes one particular realisation drawn from the probability space. For a fixed ω, $H(\cdot, \omega)$ is a deterministic spatial function (one "map" of the property over the structure); for a fixed $\mathbf{x}$, $H(\mathbf{x}, \cdot)$ is an ordinary random variable.

A **second-order** random field is fully characterised by two functions [16]:

- the **mean field** $\bar{H}(\mathbf{x}) = E[H(\mathbf{x})]$,
- the **covariance function** $C_H(\mathbf{x}, \mathbf{y}) = E\big[\big(H(\mathbf{x}) - \bar{H}(\mathbf{x})\big)\big(H(\mathbf{y}) - \bar{H}(\mathbf{y})\big)\big]$.

The mean field gives the average value at each point; the covariance function measures how strongly the values at two points $\mathbf{x}$ and $\mathbf{y}$ are linked. When both depend only on the separation $\boldsymbol{\tau} = \mathbf{x} - \mathbf{y}$, the field is called **homogeneous** (stationary).

B.2.1 Correlation function and correlation length

When a material property (e.g. Young's modulus) is modelled as a random field, nearby points tend to have similar values while distant points are more different. The **autocorrelation function** $\rho_H(\boldsymbol{\tau})$ quantifies this similarity as a function of distance τ:

$$\rho_H(\boldsymbol{\tau}) = \frac{C_H(\boldsymbol{\tau})}{\sigma_H^2}, \qquad \rho_H(\mathbf{0}) = 1, \tag{B.8}$$

where $\sigma_H^2 = C_H(\mathbf{0})$ is the field variance. $\rho = 1$ means two points have identical values; $\rho \approx 0$ means they are nearly independent.

The rate at which ρ decays is controlled by the **correlation length** $L_c > 0$. Two common models for concrete [16, 9]:

$$\rho_H(\tau) = \exp\left(-\frac{\tau^2}{L_c^2}\right) \text{ (squared-exponential)}, \qquad \rho_H(\tau) = \exp\left(-\frac{|\tau|}{L_c}\right) \text{ (exponential)}. \tag{B.9}$$

Physically, L_c represents the distance beyond which material properties become essentially independent. A small L_c means the material varies rapidly in space (highly heterogeneous); a large L_c means properties change slowly (nearly uniform). This effect is illustrated in **??**.

Remark: In this work, L_c is not treated as a parameter to be varied. A single representative value is adopted from experimental data [7] to account for spatial heterogeneity in the Monte Carlo simulations (**??**). The objective is not to study the effect of L_c itself, but to compare the structural response *with* and *without* spatial variability.

B.2.2 Random field simulation

To use a random field in a finite element model, one must *generate* realisations that respect the prescribed covariance structure, then *assign* the values to elements or integration points. Several families of methods exist [9]:

- **Matrix decomposition (Cholesky).** The covariance matrix $\mathbf{C}$ is factorised as $\mathbf{C} = \mathbf{L}\mathbf{L}^T$ and a correlated realisation is obtained as $\mathbf{h} = \mathbf{L}\mathbf{z}$, where $\mathbf{z}$ is a vector of independent standard normal variates [9]. The method is exact but scales as $\mathcal{O}(n^3)$ in the number of nodes n, making it prohibitive for large meshes.

- **Spectral methods (FFT-based).** Realisations are generated by filtering white noise in the frequency domain using the spectral density corresponding to the target covariance [9]. FFT algorithms are computationally efficient on regular grids, but the method is limited to stationary fields on rectangular domains with periodic boundary conditions.

- **Local Average Subdivision (LAS).** LAS generates the random field by successively subdividing the domain and assigning local averages consistent with the target covariance at each level of refinement [9]. The method is computationally efficient and naturally produces spatially averaged values over elements, making it well-suited for direct use in finite element models. However, it is primarily designed for regular rectangular grids.

- **Series expansion methods.** Methods such as the Karhunen–Loève (KL) expansion parametrise the random field as a finite series of deterministic spatial functions multiplied by independent random variables [16, 7]. They provide a compact representation of the spatial variability but require solving a global eigenvalue problem on the finite element mesh, whose cost increases with mesh size and targeted accuracy.

- **Turning Bands Method (TBM).** The TBM reduces the d-dimensional simulation to a set of one-dimensional stationary processes along random lines (bands) passing through the domain [12, 9]. It requires neither a global matrix factorisation nor an eigenvalue solve, and operates directly on any mesh geometry.

In this work, the TBM method, described in detail in Appendix B.2.3, is adopted because it is the method implemented in the Cast3M `ALEA` operator.

B.2.3 Turning Bands Method (`ALEA`)

Principle The Turning Bands Method (TBM), introduced by [12], simplifies the generation of a 2-D or 3-D random field by reducing it to a set of 1-D problems.

The idea (**??**): draw L lines through the domain in different directions. Along each line, simulate an independent 1-D random process. To obtain the field value at any point $\mathbf{x}$, project it onto each line and average the results:

$$Z(\mathbf{x}) \;=\; \frac{1}{\sqrt{L}} \sum_{i=1}^{L} Z_i(\mathbf{x} \cdot \mathbf{u}_i)\,, \tag{B.10}$$

where $\mathbf{u}_i$ is the direction of the i-th line and Z_i is the 1-D process along it. The 1-D processes are constructed so that the resulting field reproduces the prescribed covariance structure (Appendix B.2.1). In practice, $L = 16$ lines are sufficient for isotropic fields in two dimensions [9].

Implementation in Cast3M The TBM is natively implemented in the Cast3M `ALEA` operator, which takes as input the Gaussian parameters $(\mu_{\ln}, \sigma_{\ln})$ and the correlation length L_c, and returns the simulated field directly as nodal values on the mesh. The lognormal transformation and assignment to `MATE` follow the procedure described in Appendix B.2.4.

B.2.4 Transformation of Lognormal to Gaussian distribution

In practical finite element implementations using the Cast3M software, material parameters such as Young's modulus and permeability must remain strictly positive. However, the built-in random field operator `ALEA` generates Gaussian (normal) distributions, necessitating a mathematical transformation [16].

A **lognormal random field** $H(\mathbf{x})$ is therefore adopted: by definition, $\ln H(\mathbf{x})$ is a Gaussian random field, which guarantees $H(\mathbf{x}) > 0$ at every point [7]. The moments of $H(\mathbf{x})$ are related to those of the underlying Gaussian field $G(\mathbf{x}) \sim \mathcal{N}(\mu_{\ln},\, \sigma_{\ln}^2)$ by [16]:

$$\begin{cases} \mu_X \;=\; \exp\left(\mu_{\ln} + \dfrac{\sigma_{\ln}^2}{2}\right) \\ \sigma_X^2 \;=\; [\exp(\sigma_{\ln}^2) - 1]\exp(2\mu_{\ln} + \sigma_{\ln}^2) \end{cases} \tag{B.11}$$

where μ_X and σ_X are the physical mean and standard deviation of the material property. Inverting Equation (B.11) yields the Gaussian parameters as functions of the physical moments:

$$\sigma_{\ln}^2 = \ln\big(1 + CoV_X^2\big)\,, \qquad \mu_{\ln} = \ln(\mu_X) - \tfrac{1}{2}\,\sigma_{\ln}^2, \tag{B.12}$$

where $CoV_X = \sigma_X/\mu_X$ is the coefficient of variation.

The standard computational procedure in Cast3M `ALEA` then consists of two steps:

1. Generate a Gaussian random field $G(\mathbf{x}) \sim \mathcal{N}(\mu_{\ln},\, \sigma_{\ln}^2)$ using the Turning Band Method (see Appendix B.2.3), with parameters computed from Equation (B.12).
2. Transform pointwise to obtain the lognormal field:

$$H(\mathbf{x}) \;=\; \exp\!\big(G(\mathbf{x})\big), \tag{B.13}$$

which guarantees $H(\mathbf{x}) > 0$ and preserves the correlation structure characterised by L_c (see Appendix B.2.1).

This pointwise transformation is exact: the autocorrelation structure of the underlying Gaussian field is preserved in the lognormal field, so the spatial variability characterised by the correlation length L_c remains physically meaningful after transformation [7].

B.3 Monte Carlo simulation and LHS framework

B.3.1 Monte Carlo Principle

Monte Carlo simulation (MCS) estimates the effect of uncertain inputs by running the deterministic model multiple times, each time with a different random sample drawn from the input distributions [9]. However, standard MCS converges only as $1/\sqrt{N}$, necessitating a vast number of computationally expensive finite element runs.

B.3.2 Latin Hypercube Sampling (LHS)

To reduce the computational cost and accelerate statistical convergence, **Latin Hypercube Sampling (LHS)** is utilized as a variance-reduced alternative [**mckay1979**, **baudin2017**]. As previously introduced, the concept of the Cumulative Distribution Function (CDF) is fundamental to explaining how LHS works. In LHS, the CDF of each random variable is divided into N equiprobable strata (intervals of equal probability). Exactly one sample is drawn from each stratum by inverting the CDF, guaranteeing that the input space is covered evenly across the entire range of the probability distributions. This ensures the output statistics stabilize with markedly fewer evaluations compared to standard Monte Carlo.

B.3.3 Convergence

The probability of failure is estimated as the fraction of realisations where the limit state is violated:

$$\hat{P}_f = \frac{N_f}{n}, \tag{B.14}$$

where the hat denotes that $\hat{P}_f$ is an estimate of the true P_f, which improves as n increases, and N_f is the number of failures out of n total runs. The more runs, the more accurate the estimate—but the error only decreases as $1/\sqrt{n}$. To halve the error, one must quadruple the number of runs.

At 95% confidence, the required number of runs is [9]:

$$n \geq \frac{4\,p_f(1-p_f)}{e^2}, \tag{B.15}$$

where e is the acceptable absolute error. For a small failure probability ($p_f \approx 10^{-3}$) with 10% relative accuracy, this gives $n \approx 400{,}000$ runs. When a single THM analysis takes several minutes, this cost becomes prohibitive—motivating the Gaussian Quadrature method (Appendix B.4) as a faster alternative.

B.4 Gaussian Quadrature framework

Unlike Monte Carlo, which runs thousands of simulations, Gaussian Quadrature evaluates the model at a small number of strategically chosen points and reconstructs the output statistics from a weighted sum [**baldeweck1999**, 15]. The method has three steps, corresponding to three Cast3M operators:

1. `QUADRATU`: compute the evaluation points and weights for each random variable.
2. `PARASTAT`: run the model at each point combination and compute the first four statistical moments (mean, standard deviation, skewness, kurtosis).
3. `PROBDENS`: reconstruct the PDF from these moments and estimate P_f and β.

B.4.1 Step 1: Quadrature points and weights (`QUADRATU`)

Computing any statistical moment of the output $G = g(\mathbf{X})$ requires evaluating an integral:

$$I = \int_{\mathcal{D}} g(x)\, W(x)\, \mathrm{d}x, \tag{B.16}$$

where $W(x) = f_X(x)$ is the PDF of the input variable. Gaussian quadrature approximates this by a weighted sum over K evaluation points [**baldeweck1999**]:

$$I \approx \sum_{k=1}^{K} w_k\, g(x_k), \tag{B.17}$$

where $\{x_k\}$ are the *abscissas* (evaluation points) and $\{w_k\}$ the corresponding *weights*. The optimal abscissas are the roots of orthogonal polynomials associated with W, and a K-point scheme integrates exactly any polynomial of degree up to $2K-1$ [15].

`QUADRATU` automatically computes $\{x_k, w_k\}$ for any prescribed distribution (Normal, Lognormal, Uniform, etc.). The user provides only the distribution type and its parameters.

B.4.2 Step 2: Statistical moments (`PARASTAT`)

For N independent random variables with K points each, the quadrature is applied to each variable separately and combined as a tensor product [15]:

$$\mu_q^{(0)} \approx \sum_{k_1=1}^{K_1} \cdots \sum_{k_N=1}^{K_N} w_{k_1} \cdots w_{k_N} \left[g(x_{k_1}, \ldots, x_{k_N})\right]^q, \tag{B.18}$$

requiring K^N model evaluations in total. For example, with $N = 2$ variables and $K = 3$ points each: $3^2 = 9$ runs.

`PARASTAT` collects these evaluations and returns the first four moments: mean μ, standard deviation σ, skewness $\delta_3 = \mu_3/\sigma^3$, and kurtosis $\delta_4 = \mu_4/\sigma^4$.

B.4.3 Step 3: PDF reconstruction and reliability (`PROBDENS`)

Once the four moments are known, `PROBDENS` reconstructs the PDF using *three independent methods simultaneously* [15]:

- **Winterstein approximation:** expresses G as a Hermite polynomial transformation of a standard normal variable. Simplest but least accurate of the three methods.

- **Pearson's method:** selects the best-fit distribution family (normal, beta, gamma, etc.) from the moment values. Most accurate, but does not always provide a result.
- **Exponential of polynomials ($n = 4$):** models the PDF as $\tilde{f}_G = c\,\exp(Q)$ where Q is a polynomial fitted to the moments. Always provides accurate results.

The reconstructed Cumulative Distribution Function (CDF) serves as the fundamental basis for calculating the probability of failure P_f. Each method independently yields a probability of failure, obtained directly from the reconstructed CDF $\tilde{F}_G$ as:

$$P_f \;=\; \tilde{F}_G(0), \tag{B.19}$$

from which the reliability index follows through the general definition (Equation (B.28)). When the three reconstructions are consistent, the result can be accepted with confidence; the method is accurate for $\beta \lesssim 5$ [15].

B.4.4 Computational cost

The total number of model evaluations is K^N, which grows exponentially with N. For the present study with $N = 2$ and $K = 3$, this gives only 9 runs—compared to thousands for Monte Carlo. However, for large N (e.g. 3^{10} = 59,049 runs), the cost becomes comparable to Monte Carlo. This *curse of dimensionality* limits the method to problems with a small number of random variables ($N \lesssim$ 10–15) [15].

B.5 Sensitivity indicators and ranking methods

B.5.1 Variance decomposition

For a model $G = g(X_1, \ldots, X_N)$ with N independent random inputs, Hoeffding's decomposition [**sobol1993**] expresses G as a sum of functions of increasing dimension:

$$G \;=\; g_0 \;+\; \sum_{i=1}^{N} g_i(X_i) \;+\; \sum_{i<j} g_{ij}(X_i, X_j) \;+\; \cdots \;+\; g_{1,\ldots,N}(X_1, \ldots, X_N), \tag{B.20}$$

where $g_0 = \mathbb{E}[G]$ is a constant and each term has zero mean. Taking the variance of both sides yields:

$$\mathrm{Var}(G) \;=\; \sum_{i=1}^{N} V_i \;+\; \sum_{i<j} V_{ij} \;+\; \cdots \;+\; V_{1,\ldots,N}, \tag{B.21}$$

where $V_i = \mathrm{Var}[g_i(X_i)]$ is the first-order variance (effect of X_i alone) and $V_{ij} = \mathrm{Var}[g_{ij}(X_i, X_j)]$ is the second-order variance (interaction between X_i and X_j).

B.5.2 Variance-based sensitivity: Sobol indices

Dividing each term in Equation (B.21) by the total variance defines the *Sobol indices* [**sobol1993**]:

$$S_i \;=\; \frac{V_i}{\mathrm{Var}(G)}, \qquad S_{ij} \;=\; \frac{V_{ij}}{\mathrm{Var}(G)}, \tag{B.22}$$

where:

- S_i is the **first-order Sobol index**: fraction of the output variance due to X_i alone.
- S_{ij} is the **second-order Sobol index**: fraction due to the interaction between X_i and X_j, beyond their individual effects.

The **total-order Sobol index** captures the full effect of X_i, including all interactions:

$$S_{T_i} \;=\; S_i \;+\; \sum_{j\neq i} S_{ij} \;+\; \cdots \tag{B.23}$$

By construction, the first-order indices sum to at most one ($\sum_i S_i \leq 1$), and the total-order indices sum to at least one ($\sum_i S_{T_i} \geq 1$). When the model is additive (no interactions), $S_{ij} = 0$ for all pairs and $S_i = S_{T_i}$.

The first-order index S_i can be expressed as a conditional variance:

$$S_i \;=\; \frac{\mathrm{Var}_{X_i}\left[\mathbb{E}_{X_{\sim i}}(G \mid X_i)\right]}{\mathrm{Var}(G)}, \tag{B.24}$$

where $X_{\sim i}$ denotes all variables except X_i. Intuitively, $\mathbb{E}(G \mid X_i)$ averages out all other variables: if this conditional mean varies a lot with X_i, then X_i is influential.

A variable with a large S_i (or S_{T_i}) is a *dominant parameter*: it drives most of the output uncertainty and should be characterised with care. A variable with a small index could be treated as deterministic without significant loss of accuracy.

B.5.3 Practical sensitivity approximation

Computing exact Sobol indices requires a large number of model evaluations. A simpler approach, used by the the expansion of Englishquadrature method (Appendix B.4), estimates the contribution of each variable one at a time:

1. Fix all variables at their mean except X_i.
2. Run the model at each quadrature point of X_i to obtain the marginal variance $\hat{\sigma}_i^2$.
3. Repeat for every variable.

The **contribution** of X_i to the total output variability is then:

$$C_i = \frac{\hat{\sigma}_i^2}{\sum_{j=1}^{N} \hat{\sigma}_j^2} \times 100\,\% \approx S_i \quad \text{when } S_{ij} \approx 0. \tag{B.25}$$

A variable with $C_i \approx 0\,\%$ has negligible influence and could be treated as deterministic. This approximation is valid when variables are uncorrelated and interaction effects are small. The detailed results are presented in **??**.

B.6 Probability of failure and reliability index

B.6.1 Limit state function

In structural reliability, a **performance function** (or limit-state function) $G(\mathbf{X})$ explicitly separates the structural state into two domains:

$$G(\mathbf{X}) > 0 \Rightarrow \text{safety domain } D_s, \qquad G(\mathbf{X}) \leq 0 \Rightarrow \text{failure domain } D_f. \tag{B.26}$$

For a typical structure evaluated against a specific load S and a critical resistance R, the function reads $G(R, S) = R - S$ [2].

B.6.2 Probability of failure and reliability index β

The **probability of failure** is the integral of the joint probability density function over the failure domain [2]:

$$P_f = \text{Prob}\big[G(\mathbf{X}) \leq 0\big] = \int_{G(\mathbf{X})\leq 0} f_{\mathbf{X}}(\mathbf{x})\,\mathrm{d}\mathbf{x}. \tag{B.27}$$

However, when dealing with complex non-linear finite element models, this multidimensional integral cannot be solved analytically and must be estimated through numerical approximation methods such as Monte Carlo simulation or Kernel Density Estimation (KDE).

In modern engineering codes, P_f is frequently mapped to the **generalized reliability index** β via the inverse of the standard normal cumulative distribution function Φ^{-1}:

$$\beta = -\Phi^{-1}(P_f). \tag{B.28}$$

B.7 Reference parameters for the simplified poro-mechanical model

To evaluate the effect of spatial heterogeneity in Chapter 5, a simplified isothermal poro-mechanical model is implemented in CAST3M. This section details the specific constitutive laws and material parameters used for that test case.

B.7.1 Constitutive relationships

The hygral behaviour is governed by the van Genuchten model [**vangenuchten1980**]. The degree of saturation S_l and the liquid relative permeability k_{rl} are functions of the capillary pressure P_c:

$$S_l(P_c) = \left[1 + \left(\frac{P_c}{a_1}\right)^{\frac{b_1}{b_1-1}}\right]^{-\frac{1}{b_1}} \tag{B.29}$$

$$k_{rl}(P_c) = \sqrt{S_l}\left[1 - \left(1 - S_l^{b_1}\right)^{\frac{1}{b_1}}\right]^2 \tag{B.30}$$

where a_1 and b_1 are material parameters. The effective hydraulic conductivity K_{eff} entering the flow matrix is:

$$K_{\text{eff}}(P_c) = \frac{k_{rl}(P_c)\,\rho_l\,K_0}{\mu_l} \tag{B.31}$$

B.7.2 Geometry, mesh, and material parameters

The modelled specimen is a $10 \times 10 \times 10$ cm cube. By exploiting symmetry, only 1/8 of the domain is meshed using 3D 8-node hexahedral elements (`CUB8`). The mesh is refined near the exposed surfaces, transitioning from a coarse element size of 10 mm down to 2 mm. The transient simulation spans 120 days with a time step of 1 day.

The deterministic reference parameters fed into the model are summarised in Table B.1.

For the stochastic analyses (C1, C2, C3), the intrinsic permeability K_0 is treated as a random variable or a spatial random field centered around the mean value of 1.0×10^{-20} m^2.

Table B.1: Reference parameters for the simplified isothermal poro-mechanical model.

Parameter	Symbol	Value	Unit
Hygral properties			
Liquid water density	ρ_l	1000	kg/m^3
Dynamic viscosity	μ_l	1.0×10^{-3}	Pa · s
Porosity	ϕ	0.20	–
Mean intrinsic permeability	K_0	1.0×10^{-20}	m^2
van Genuchten parameter	a_1	20	MPa
van Genuchten parameter	b_1	2.0	–
Mechanical properties			
Young's modulus	E	30	GPa
Poisson's ratio	ν	0.20	–
Biot coefficient	α	0.60	–
Initial and boundary conditions			
Temperature	T	20	°C
Initial relative humidity	$\mathrm{RH}_{\mathrm{ini}}$	0.90	–
External relative humidity	$\mathrm{RH}_{\mathrm{ext}}$	0.50	–

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EDUCATION

Université Grenoble Alpes **Sep 2025 - Present**

Master 2: Geomechanics, Civil Engineering & Risks *Grenoble, France*

- **M2 Semester 1 Grade:** 13.47/20 (Admis).
- **Key Coursework:** Numerical Methods for Nonlinear Mechanics, Durability and Vulnerability of Structures (15.8/20), Quantitative Image Analysis for Mechanics (16/20).
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- **GPA:** 8.06/10 (3.5/4.0) | **Rank:** 3/12 (Excellence Program).
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RESEARCH & PROFESSIONAL EXPERIENCE

M2 Research Intern / Thesis Student **Feb 2026 - Present**

3SR Laboratory, Université Grenoble Alpes *Grenoble, France*

- **Repository:** github.com/tankiet1907/cast3m-m2internship
- Investigating the thermo-hydro-mechanical (THM) behavior of concrete under severe accident conditions using a poro-mechanical approach.
- Developing and implementing numerical behavior models within the **Cast3M** finite element code using Gibiane.
- Utilizing probabilistic and statistical methods to conduct sensitivity analyses and optimize THM coupling parameters.
- Correlating numerical simulations with experimental data to validate predictive models for structural integrity.

Structure Engineer (Full Time) **Aug 2024 - Aug 2025**

GBC Engineers Vietnam *Ho Chi Minh City, Vietnam*

- Prepared Finite Element Models (FEM) and structural models for rigorous mechanical analysis.
- Delivered structural calculations, analysis reports, and correspondence ensuring compliance with international standards (Eurocode, TCVN).

Site Engineer (Internship) **Jun 2023 - Aug 2023**
Unicons Construction (Member of Coteccons Group) *Binh Duong, Vietnam*

- **Project:** LEGO Manufacturing Vietnam (Phase 1) – Moulding Building.
- Supervised structural fire protection implementations (Fire Protection Boards, Intumescent Paint, Concrete-encasement) to meet rigorous 120-minute Fire Protection Ratings.
- Optimized the execution schedule by transitioning the concrete encasement casting for columns from vertical to horizontal pouring.
- Conducted Quality Control per EC2: inspected steel frame erection via total station, verified tightening forces for Snug and Bearing bolts, and supervised Sika grouting and composite decking.

PROJECTS

Scientific Research: Fire-Resistant Steel Structures **Sep 2023 - Dec 2023**
Ho Chi Minh City University of Technology *Ho Chi Minh City, Vietnam*

- Evaluated thermo-mechanical properties of steel at high temperatures per EUROCODE specifications.
- Designed and verified fire-resistant structural elements (columns and beams) using advanced material models.

CERTIFICATIONS & AWARDS

- **Third Prize,** 36th **Loa Thanh Award (2024):** National award for the most prestigious graduation projects in civil engineering.
- **Third Prize in Physics (2018):** City Level Excellent Student Competition (Ho Chi Minh City).
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SKILLS & COMPETENCIES

Technical: Cast3M (Advanced), Finite Element Analysis (FEA), THM Coupling, Probabilistic Sensitivity Analysis.

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EXTRACURRICULAR ACTIVITIES

- **Main Vocal:** Tet Festival of the Vietnamese Community in Grenoble. 2026
- **Main Vocal:** PFIEV Freshman Welcome Party (2023) and Journée de la Culture Française (2022). 2022 - 2023
- **Volunteer:** Supporting COVID-19 prevention work.
- **Security Support:** Admissions Consulting Day (2022) and University Entrance Exam Support Program (2020). 2020 - 2022.